语音增强前沿进展

从预测式估计迈向生成式建模

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- 上海交通大学博士、博士后
- 研究方向：语音增强、语音识别与语音大模型
- 关注预测式、无监督与生成式语音增强

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学习目标与时间安排

◎ 本教程希望回答四个问题

1. 直接估计一个结果与学习语音分布，有何不同？
2. VAE、GAN、扩散和流匹配如何用于增强？
3. 预测与生成如何互补，又如何避免生成偏差？
4. 怎样综合评价保真度、听感与计算成本？

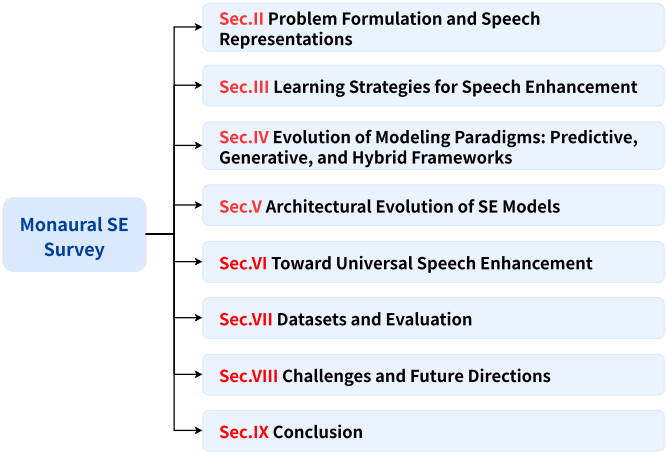
面向语音研究者与工程师

预备知识：基本概率论与神经网络。



时间安排：约 90 分钟

章节	主题	分钟
II	问题定义与语音表征	8
III	语音增强模型的学习策略	7
IV	建模范式的演进：预测、生成与混合	30
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V	语音增强模型的架构演进	10
VI	迈向通用语音增强	10
VII	数据集与评测	10
VIII	挑战与未来方向	5
–	互动问答	5



论文贡献：

- ✓ 从问题定义、语音表征、学习策略与建模范式等互补维度统一组织单通道语音增强
- ✓ 系统梳理预测式、生成式与混合式代表方法及其架构设计，含多分支与面向效率的架构
- ✓ 综述通用语音增强、代表性数据集与评测协议，并在同一基准上跨范式比较代表方法
- ✓ 在此基础上讨论高效推断、可靠评测、域偏移鲁棒性与语音基座模型集成等开放挑战

W. Jiang, W. Zhang, J. Zhang, F. Wen, L. Xie, and K. Yu, "A unified survey of monaural speech enhancement: Predictive, generative, and beyond," *IEEE TASLP*, **Accepted**.

本教程的术语约定

术语	本教程中的含义
预测式 / Predictive	直接输出纯净语音的点估计或掩码
生成式 / Generative	学习语音先验或条件分布，并据此生成增强结果
混合式 / Hybrid	结合预测估计与生成建模，包括模块组合与单模型联合学习
得分函数 / Score	对数概率密度对信号的梯度，刻画分布的局部方向
流匹配 / Flow Matching	学习把基分布输运到目标分布的速度场
潜变量 / latent variable	VAE 等模型中承载语音变化的随机变量，常记作 $\mathbf{z}$
通用语音增强 / Universal SE	面向多种及复合退化的语音恢复
退化语音 / Degraded speech	泛指受到噪声、混响、编码失真等各种退化影响的语音
语音 token	离散语音单元；语义 token 与声学 token 保留的信息不同

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3. 建模范式的演进：预测式、生成式与混合框架
4. 语音增强模型的架构演进
5. 迈向通用语音增强
6. 数据集与评测
7. 挑战与未来方向

从应用需求理解语音增强



通信与会议

嘈杂环境中听清对方；抑制噪声，同时保持自然语音与对话节奏。

关注：可懂度、听感、时延



助听与耳机

设备算力和电池有限；弱语音不能被过度抑制，处理延迟也不能过大。

关注：低失真、功耗、延迟



语音交互前端

为识别与说话人系统提供输入；降噪后仍要保留词句、音色和身份信息。

关注：内容、声纹、下游任务

根据应用场景的需求，选择合适的方法：

- 实时通话优先保证对话流畅
- 离线处理允许非因果系统
- 识别前端需更小的语义失真

[1] P. C. Loizou, "Speech Enhancement: Theory and Practice," CRC Press, 2013.
[2] E. Vincent et al., "Audio Source Separation and Speech Enhancement," John Wiley & Sons, 2018.

语音增强是指从退化观测语音中恢复干净的语音信号。令 $\mathbf{x} \in \mathbb{R}^N$ 表示分帧后的干净语音信号，$\mathbf{y} \in \mathbb{R}^N$ 表示观测到的退化信号。一般而言，退化和恢复过程可写为：

$$\mathbf{y} = d_\eta(\mathbf{x}) + \mathbf{n}, \qquad \hat{\mathbf{x}} = \mathcal{F}(\mathbf{y})$$

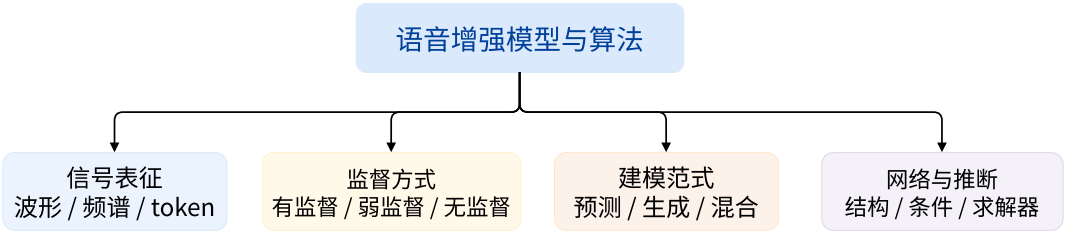
- $\mathbf{x}$：纯净语音；$\mathbf{y}$：观测语音；$\mathbf{n}$：加性噪声或其他干扰
- d_η：混响、带宽限制、设备或编码退化
- $\mathcal{F}$：恢复函数，可以为确定性的、或者具有一定随机性的
- 从信号处理的角度看，仅凭单个退化观测恢复干净语音本身通常是欠定的；实际系统需要借助先验信息、结构约束或数据驱动模型，才能得到稳定且合理的估计

讨论

当高频细节已丢失，“逼真”和“与原信号完全一致”能否同时保证？

[3] J.-M. Lemercier et al., “Analysing diffusion-based generative approaches versus discriminative approaches for speech restoration,” ICASSP 2023.
[4] J.-M. Lemercier et al., “Diffusion models for audio restoration: A review,” IEEE Signal Processing Magazine, 2024.

语音增强模型与算法的四个维度



四个维度从不同的侧面刻画语音增强算法

以信号表征、建模范式、监督方式进行说明：

表征与范式：DCCRN 与 SGMSE+ 均用复数 STFT；前者预测掩码，后者学习条件得分并迭代采样

训练监督：DCCRN 为有监督学习；RemixIT 为弱监督学习，以教师模型估计重混训练学生模型

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.
[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.
[7] E. Tzinis et al., "RemixIT: Continual self-training of speech enhancement models via bootstrapped remixing," IEEE J. Sel. Top. Signal Process., 2022.

语音表征概览：类别、代表方法与关键特性

Name	Year	Architecture	Representations
Classic			
STFT	<1980s	STFT	Complex spectrogram
Mel[8]	<1980s	Mel filterbank	Mel spectrogram
Learnable Transform			
Conv-TasNet[9]	2018–19	1D CNN	Real-valued feature
Continuous Embedding			
VAE[10]	2014	Encoder–Decoder	Continuous latent var.
HuBERT[11]	2021	CNN+Transformer	Contextual repr.
WavLM[12]	2022	CNN+Transformer	Contextual repr.
Whisper[13]	2023	CNN+Transformer	Contextual repr.
Discrete Token			
VQ-VAE[14]	2017	CNN+VQ	Discrete latent codes
EnCodec[15]	2023	CNN+LSTM+RVQ	Acoustic tokens

- 四类表征：经典固定变换、可学习变换、连续嵌入、离散 token
- 经典的 STFT 为当前的主流方法：仅建模幅度可简化学习，但会丢失相位信息；复数谱建模可同时恢复幅度与相位
- Mel 域特征表示模拟人耳感知特性：突出关键频率、降低冗余，常用于生成式框架
- Continuous Embedding 和 Discrete Token 表征可用于端到端建模，尤其适合语音生成和语音大模型

[10] D. P. Kingma et al., "Auto-encoding variational Bayes," ICLR 2014.
[11] W.-N. Hsu et al., "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," IEEE/ACM TASLP, 2021.

代表方法分类：建模范式 × 学习策略

	Supervised	Weakly Supervised	Unsupervised
Predictive	Conv-TasNet[9], DCCRN[5], TF-GridNet[16], MP-SENet[17]	Noise2Noise[18], RemixIT[7], IDR-SE[19], SelfSE[20]	MetricGAN-U[21], DOTN[22], UnSE[23]
Generative	SEGAN[24], SGMSE+[6], FlowSE-Vocos[25], SenSE[26]	–	VAE–NMF[27], RVAE[28], DVAE–VEM[29]
Hybrid	StoRM[30], Diffiner[31], PGUSE[32], HyFlowSE[33], SIPS[34]	–	–

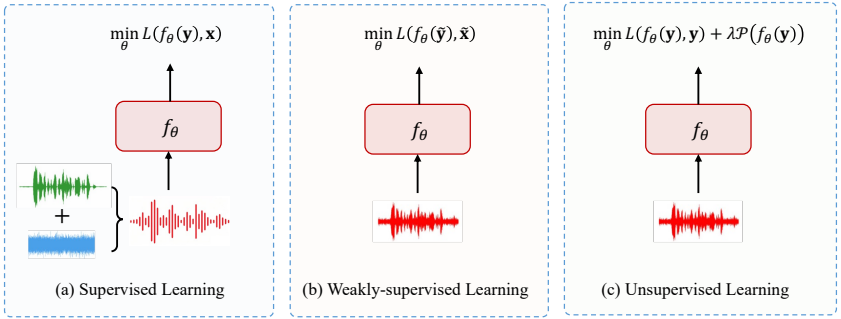
- 分类是根据学习范式和学习策略，而非方法或者网络名称：对抗训练为点估计器时归类预测式（如 MetricGAN），完整语音分布建模为生成式（如 VAE-NMF），显式结合预测估计与生成推断为混合式（如 StoRM）
- 有监督学习仍然是当前主流学习策略，弱监督与无监督主要解决训练数据难以获取的问题
- 无监督生成式方法通常依赖语音先验＋变分或后验推断（如 VAE–NMF）

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.
[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.

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语音增强模型的学习策略：有监督、弱监督与无监督



监督学习

使用配对的带噪与纯净语音；

缺点：合成的训练数据与真实场景不匹配

弱监督学习

使用教师模型输出、带噪代理 (Noisy Proxy) 或其他不精确目标；

缺点：目标偏差与误差积累

无监督学习

不使用逐样本的纯净或伪纯净目标，借助语音先验、一致性约束或质量评价器；

缺点：与有监督相比存在差距

监督方式描述训练信息，与建模范式相对独立

语音增强模型的学习策略：典型代表性方法对比

Method	Year	Category	Teach.	Noisy	Clean	Noise
OMLSA[35]	2001	Classic	✗	Real	✗	✗
DNN[36]	2015	Supervised	✗	Simu	✓	✗
CRN[37]	2018		✗	Simu	✓	✗
DCCRN[5]	2020		✗	Simu	✓	✗
TF-GridNet[16]	2023		✗	Simu	✓	✗
NyTT[38]	2021	Weakly sup.	✗	Real	✗	✓
Noise2Noise[18]	2021		✗	Simu	✗	✓
RemixIT[7]	2022		✓	Real	✗	✗
IDR-SE[19]	2024		✗	Real	✗	✓
SelfSE[20]	2026		✗	Real	✗	✓
DOTN[22]	2021	Unsupervised	✗	Simu	✓	✗
MetricGAN-U[21]	2022		✗	Real	✗	✗
DVAE-VEM[29]	2022		✗	Real	✓	✗
UnSE[23]	2024		✗	Real	✓	✗

- 基于信号处理的经典方法仅需要带噪语音，无需模型训练
- 有监督：依赖成对带噪–纯净数据
- 弱监督：显式但不完整的信息——噪声代理目标（NyTT、Noise2Noise）、教师模型估计（RemixIT）、迭代精化（IDR-SE、SelfSE）
- 无监督：不用配对或代理目标——分布匹配/变换一致性（DOTN、UnSE）、潜变量推断（DVAE-VEM）、非侵入质量目标（MetricGAN-U）

[7] E. Tzinis et al., "RemixIT: Continual self-training of speech enhancement models via bootstrapped remixing," IEEE J. Sel. Top. Signal Process., 2022.
[20] W. Jiang et al., "SelfSE: Self-supervised speech enhancement via noisy speech refinement," IEEE TASLP, 2026.

弱监督学习 (Weakly Supervised Learning) 不要求成对纯净目标，适用于无法获取纯净参考的真实场景：

$$\min_{\theta} \mathbb{E}\big[L\big(f_\theta(\tilde{\mathbf{y}}), \tilde{\mathbf{x}}\big)\big], \qquad \tilde{\mathbf{x}} = \pi(\mathbf{y})\ \text{（由带噪观测构造伪目标）}$$

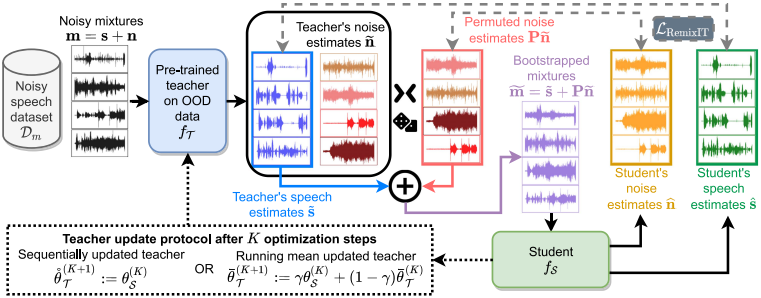
- 伪目标显式但不完整、不精确：带噪代理（NyTT）、教师模型估计（RemixIT）、自我迭代（IDR-SE、SelfSE）
- 从不完美目标（如：估计的语音）或部分任务信息中学习（如：文本信息）
- 伪目标可靠性与训练稳定性是主要挑战（模型容易训练崩掉）

与有监督学习的关系

损失函数与有监督一致，只是目标来自带噪观测本身或外部模型，而非配对纯净语音

[7] E. Tzinis et al., "RemixIT: Continual self-training of speech enhancement models via bootstrapped remixing," IEEE J. Sel. Top. Signal Process., 2022.
[20] W. Jiang et al., "SelfSE: Self-supervised speech enhancement via noisy speech refinement," IEEE TASLP, 2026.

弱监督学习：RemixIT



图片来源：RemixIT，Fig. 1

- 教师模型估计语音与噪声；打乱噪声后重混输入
- 学生模型以教师模型估计为目标，并可用学生模型权重更新教师模型

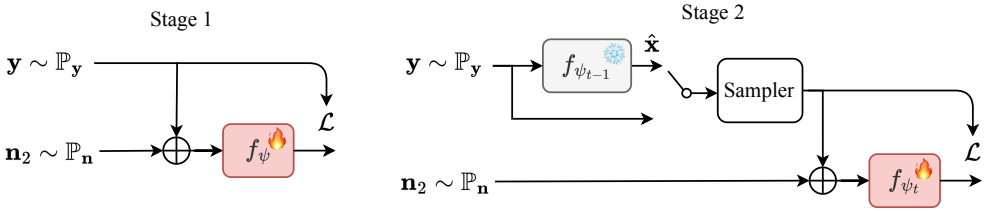
DNS test: unsupervised RemixIT (paper Table I)

Model (depth U)	Param.(M)	SI-SDR↑	PESQ↑	STOI↑
Teacher ($U = 8$)	0.79	14.8	2.15	0.940
Student ($U = 8$)	0.56	15.5	2.27	0.947
Student ($8 \to 16 \to 32$)	0.73	16.0	2.34	0.952

Teacher: LFSD MixIT; student: DNS mixtures.

[7] E. Tzinis et al., "RemixIT: Continual self-training of speech enhancement models via bootstrapped remixing," IEEE J. Sel. Top. Signal Process., 2022.

弱监督学习：SelfSE



图片来源：SelfSE，Fig. 1

两阶段方法：

Stage 1 ｜**初始化**以 $(\mathbf{y}+\mathbf{n}_2, \mathbf{y})$ 训练：输入额外加噪，目标为原带噪语音。

Stage 2 ｜**更新**上一轮估计 $\hat{\mathbf{x}}_{t-1} = f_{\psi_{t-1}}(\mathbf{y})$ 参与构造新训练目标。

Stage 2：按概率采样训练数据对

$$(\widetilde{\mathbf{y}}, \widetilde{\mathbf{x}}) = \begin{cases} (\hat{\mathbf{x}}_{t-1} + \mathbf{n}_2, \hat{\mathbf{x}}_{t-1}), & \text{概率} p, \\ (\mathbf{y} + \mathbf{n}_2, \mathbf{y}), & \text{概率} 1-p. \end{cases}$$

$p = 0$ 退化为 NyTT；$p = 1$ 退化为 IDR-SE；中间值保留原带噪目标作为锚点。

[20] W. Jiang et al., "SelfSE: Self-supervised speech enhancement via noisy speech refinement," IEEE TASLP, 2026.

语音增强模型的学习策略：无监督学习

无监督学习在没有成对的含噪-干净数据、含噪代理目标或任务相关监督信息的情况下训练模型：

$$\min_{\theta} \mathbb{E}_{\mathbf{y}} \left[L(f_\theta(\mathbf{y}), \mathbf{y}) + \lambda \mathcal{P}(f_\theta(\mathbf{y})) \right]$$

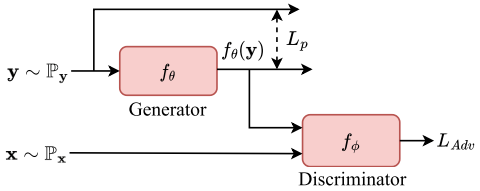
式中的约束项 $P(\cdot)$ 一般通过以下方法实现：

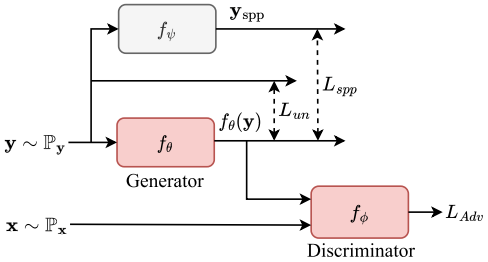
- 输入一致性：约束输出尽可能保留输入信息
- 语音质量先验分布：使输出分布逼近纯净语音

优化目标的构建：分布匹配或变换一致性（如 DOTN 和 UnSE），带有干净语音先验的隐变量推断（如 DVAE-VEM），或预训练的非侵入式质量目标（如 MetricGAN-U 和 MOS-GAN）。

[23] W. Jiang et al., "Unsupervised speech enhancement using optimal transport and speech presence probability," IEEE/ACM TASLP, 2024.
[21] S.-W. Fu et al., "MetricGAN-U: Unsupervised speech enhancement/dereverberation based only on noisy/reverberated speech," ICASSP 2022.
[27] Y. Bando et al., "Statistical speech enhancement based on probabilistic integration of variational autoencoder and non-negative matrix factorization," ICASSP 2018.

无监督学习：UnSE/UnSE+





图片来源：UnSE / UnSE+，Fig. 1

- 带噪 $\mathbf{y} \sim P_\mathbf{y}$ 与纯净 $\mathbf{x} \sim P_\mathbf{x}$ 为随机采样
- 生成器估计时频掩码；L_p 保真项约束输入，WGAN 判别器促使输出分布逼近纯净语音分布

$$L_G = \alpha_p \|f_\theta(\mathbf{y}) - \mathbf{y}\|_p^p + L_{\text{Adv}}(G; D)$$

- UnSE+：增加语音存在概率（SPP）与 STFT 域损失

VCTK+DEMAND test set

Method	PESQ↑	SI-SNR (dB)↑	eSTOI↑	CSIG↑
Noisy	1.97	8.45	0.79	3.35
UnSE[23]	2.45	15.96	0.80	**3.69**
UnSE+	**2.48**	**17.40**	**0.81**	3.31

[23] W. Jiang et al., "Unsupervised speech enhancement using optimal transport and speech presence probability," IEEE/ACM TASLP, 2024.

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6 数据集与评测

7 挑战与未来方向

预测式、生成式与混合式

预测式、生成式和混合式建模范式为语音增强提供了互补的选择：

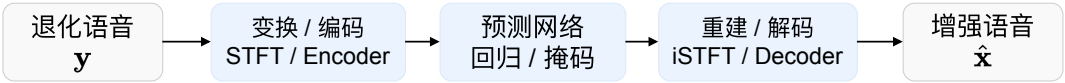
- 预测式方法直接估计干净语音，因其简单、高效且在匹配条件下性能较强而被广泛采用
- 生成式方法通过建模干净语音分布或给定含噪观测的条件分布，将语音增强表述为概率推断，从而能得到多个合理的重建结果
- 混合框架结合这两种范式的互补特性，在输入保真度、感知质量、鲁棒性和计算效率之间进行权衡

范式 / 方法	神经网络学习了什么？	推理过程有什么的特点？
预测式 / DCCRN[5]	纯净语音或掩码的点估计	单次前向，直接重建
生成式 / SGMSE+[6]	给定观测的语音，学习得分函数	迭代更新复数谱，生成增强结果
混合式 / HyFlowSE[33]	在同模型内联合生成与重建目标	很少的步数求解 ODE 得到输出

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.
[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.
[33] Y. Zhang et al., "HyFlowSE: Hybrid end-to-end flow-matching speech enhancement via generative-discriminative learning," ICASSP 2026.

预测式语音增强方法：直接映射与掩码估计

简化的流程图：



用简单的公式表示：

$$\widehat{X} = f_\theta(y) \qquad \text{或} \qquad \widehat{X} = M \odot Y = f_\theta(y) \odot \widehat{Y}$$

直接映射/回归

- 直接估计纯净语音谱或波形
- 输出形式灵活，但需控制失真

掩码 (mask) 估计

- 只学习掩码值，模型更稳定
- 依赖信号的表征和掩码定义

[36] Y. Xu et al., "A regression approach to speech enhancement based on deep neural networks," IEEE/ACM TASLP, 2015.
[39] Y. Wang et al., "On training targets for supervised speech separation," IEEE/ACM TASLP, 2014.
[40] D. S. Williamson et al., "Complex ratio masking for monaural speech separation," IEEE/ACM TASLP, 2016.

损失函数：指导模型训练的目标

损失函数决定模型“优化什么”，实际系统通常组合使用多类损失：

目标	主要约束	特点
波形 MSE / SI-SNR	信号级重建：逐点误差或尺度不变信噪比	对相位、延迟与音量对齐敏感，MSE 易导致过度平滑
幅度 / 复数谱损失	频谱结构：约束 STFT 幅度或复数谱	只用幅度会损失相位，不同的复数分解可以相互补偿
多尺度 STFT 损失	多时间尺度结构：兼顾瞬态与稳态成分	源于声码器，有额外计算上的开销
感知 / 指标驱动目标	听感或预测指标：如 PESQ、STOI、MOS 预测等	可能被模型“刷分”，需要跟其他指标一起优化

多目标权衡：调整权重后，需检查未参与优化的指标；训练目标的分数不足以代表最终质量

[41] J. Le Roux et al., “SDR–half-baked or well done?,” ICASSP 2019.
[42] J. Engel et al., “DDSP: Differentiable digital signal processing,” ICLR 2020.
[43] Z.-Q. Wang et al., “On the compensation between magnitude and phase in speech separation,” IEEE Signal Process. Lett., 2021.

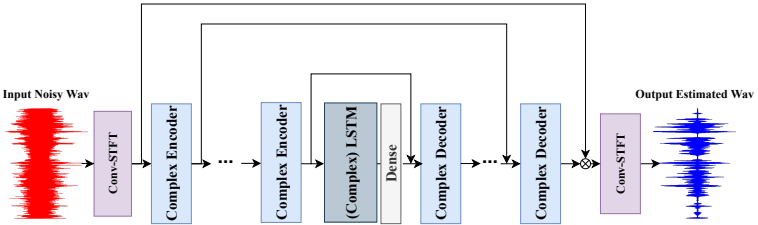
代表性预测式语音增强方法对比

Method	Year	Domain	Formulation	Loss
DNN[36]	2015	T-F	Regression	MSE
CRN[37]	2018	T-F	Regression	MSE
Conv-TasNet[9]	2019	Time	Masking	SI-SNR
DCCRN[5]	2020	T-F	Masking	SI-SNR
DEMUCS[44]	2020	Time	Regression	Combined
FullSubNet[45]	2021	T-F	Masking	MSE
DPT-FSNet[46]	2022	T-F	Masking	Combined
FRCRN[47]	2022	T-F	Masking	Combined
TF-GridNet[16]	2023	T-F	Regression	Combined
CMGAN[48]	2024	T-F	Masking	Combined
SEMamba[49]	2024	T-F	Masking	Combined
MP-SENet[17]	2025	T-F	Masking	Combined
ZipEnhancer[50]	2025	T-F	Regression	Combined

- 模型输入：时域方法（Conv-TasNet、DEMUCS）经学习编码器处理波形，时频域（T-F）仍是主流方法
- 模型输出：直接回归与掩码估计两类并存，均被广泛使用
- 损失函数：从 MSE/SI-SNR 演进为多损失函数的组合，兼顾信号保真与感知质量，包括最新的方法（SEMamba、ZipEnhancer）

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.
[17] Y.-X. Lu et al., "Explicit estimation of magnitude and phase spectra in parallel for high-quality speech enhancement," Neural Networks, 2025.

预测式语音增强方法：DCCRN



图片来源：DCCRN，Fig. 1

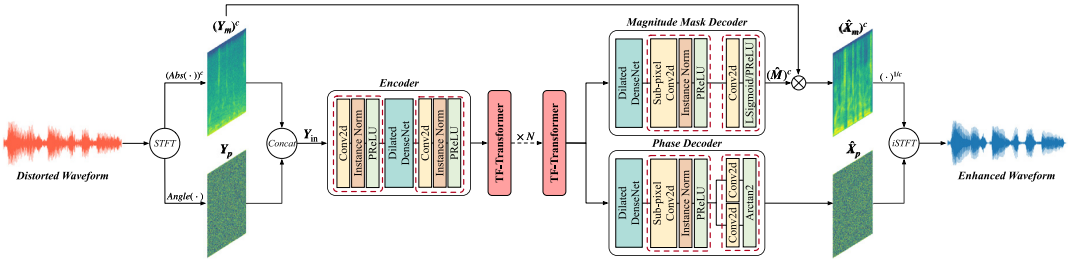
- 结合 DCUNET 与 CRN，并用 LSTM 建模时序，在低参数、低计算下实现相位感知增强
- 大幅优于 CRN，并以约 $1/6$ 计算量、相近参数达到与 DCUNET 相当的性能
- 3.7M 参数模型在 DNS 2020 P.808 评测中获实时赛道 MOS 第一、非实时赛道第二

WSJ0 数据集

Model	Param.(M)	0 dB	10 dB	Avg.
CRN[37]	6.1	2.850	3.374	3.329
DCCRN-E	3.7	2.859	3.492	3.433
DCCRN-CL	3.7	2.972	3.559	3.498
DCUNET	3.6	2.971	3.556	3.500

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.

预测式语音增强方法：MP-SENet



图片来源：MP-SENet，Fig. 1

- 带噪幅度谱与相位谱共同编码
- T-F Transformer 交替建模时间与频率依赖
- 幅度掩码和相位并行估计，iSTFT 重建
- 使用了相位上的抗卷绕损失函数

VoiceBank + DEMAND ablation; PD: phase distance (°)

Model / ablation	Param.(M)	PESQ↑	PD↓	SI-SDR↑	ViSQOL↑
MP-SENet[17]	2.26	3.60	7.28	19.47	4.34
w/ Conformer	1.71	3.56	7.35	19.38	4.30
Magnitude only	1.99	3.44	8.00	18.37	4.28
Complex only	2.26	3.58	7.56	18.57	4.28
w/o Mag. loss	2.26	3.64	7.97	14.76	3.85
w/o Pha. loss	2.26	3.52	7.65	19.25	4.32
w/o Metric Disc.	2.26	3.44	7.38	20.05	4.25

[17] Y.-X. Lu et al., "Explicit estimation of magnitude and phase spectra in parallel for high-quality speech enhancement," Neural Networks, 2025.

生成式语音增强方法的两条基本路线

预测试方法是点估计，生成式方法刻画的是干净语音上的概率分布，有两种形式：

条件分布

$$p_\theta(\mathbf{x} \mid \mathbf{y}), \qquad \mathbf{x} = G_\theta(\mathbf{y}, \mathbf{z})$$

语音先验与退化似然

$$p_\theta(\mathbf{x} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{x}) p_\theta(\mathbf{x})$$

- 条件分布方法：$\mathbf{z}$ 是从基分布 $p(\mathbf{z})$ 中采样的隐随机向量，例如 SEGAN 、SGMSE+、FlowSE 等等
- 基于先验的方法：学习干净语音先验 $p_\theta(\mathbf{x})$，并将其与退化似然 $p(\mathbf{y}|\mathbf{x})$ 结合，以推断后验 $p_\theta(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x}) p_\theta(\mathbf{x})$，例如：VAE-NMF

[27] Y. Bando et al., "Statistical speech enhancement based on probabilistic integration of variational autoencoder and non-negative matrix factorization," ICASSP 2018.

[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.

[51] S. Leglaive et al., "A variance modeling framework based on variational autoencoders for speech enhancement," IEEE Int. Workshop Mach. Learn. Signal Process. (MLSP) 2018.

生成式语音增强方法：不同的实现范式

不同生成范式目标相同一致（都刻画干净语音分布），但参数化与实现方式不同：

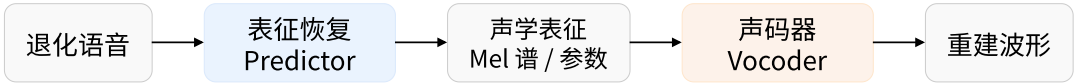
范式	如何表示分布	训练依据	常见推断
VAE[10]	潜变量、解码分布	重建似然与 KL	变分 / 后验推断
GAN	隐式生成分布	判别器反馈	生成器前向
Normalizing Flow	可逆变换与似然	变量变换后的似然	变换 / 反变换
扩散 / 得分模型	加噪过程与得分函数	去噪 / 得分匹配	SDE / ODE 采样
Flow Matching	连续概率输运	速度场学习	ODE 积分
序列生成	token 条件概率	交叉熵 / 掩码目标	自回归 / 掩码生成

都属于生成式建模，但分布表示、训练目标和采样方式不同

[52] I. J. Goodfellow et al., "Generative adversarial nets," NIPS 2014.
[53] J. Ho et al., "Denoising diffusion probabilistic models," NeurIPS 2020.
[54] Y. Lipman et al., "Flow matching for generative modeling," ICLR 2023.

生成式语音增强方法：重合成与基于声码器

基于声码器的方法有些是直接重合成，有些是联合优化声码器与增强模型：



- **输入**：模型接收“退化语音”作为输入信号
- **表征恢复**：通过 **Predictor** 模块，从退化语音中预测并恢复出目标语音的中间特征
- **声学表征**：预测器的输出为“Mel 谱”或相关的声学参数
- **波形生成**：将恢复出的声学表征送入 **Vocoder（声码器）**
- **输出**：声码器最终合成并输出“重建波形”（即增强后的语音）

近两年的通用语音增强、融合 LLM 的语音增强等方法，很多都采用类似的思路，只是这个 Predictor 换成了更强的模型

[55] S. Maiti et al., “Speech denoising by parametric resynthesis,” ICASSP 2019.
[56] S. Maiti et al., “Parametric resynthesis with neural vocoders,” WASPAA 2019.
[57] Z. Du et al., “A joint framework of denoising autoencoder and generative vocoder for monaural speech enhancement,” IEEE/ACM TASLP, 2020.

代表性重合成与声码器语音增强方法

Method	Year	Vocoder	Predictor	Input	Output
PR[55]	2019	WORLD	DNN / LSTM	Mel-spec.	Param.
Neural-PR[56]	2019	WaveNet / WaveGlow	BiLSTM	Mel-spec.	Mel-spec.
CRN-Flow[57]	2020	FloWaveNet	CRN	Mel-spec.	Mel-spec.
SE-Flow[58]	2021	WaveGlow	–	Waveform	Waveform
LA-VocE[59]	2023	HiFi-GAN	Transformer	Mel-spec.	Mel-spec.
HiFi++[60]	2023	HiFi-GAN	SpectralUNet	Mel-spec.	Waveform
DashengSE[61]	2025	Vocos	Multi	Waveform	Waveform
DDSP-SE[62]	2025	DDSP	Conformer	Mel-spec.	Param.

- PR 以 DNN/LSTM 从 Mel 谱预测 WORLD 声码器参数；神经声码器成熟后联合优化增强与波形生成（CRN-Flow、SE-Flow）
- 近期转向高保真与高效设计：LA-VocE、HiFi++、DashengSE、DDSP-SE
- 生成式还是预测试？要看特征预测器或者声码器是否为生成式，不能仅看名字（带 GAN 的声码器不一定是生成式，如 HiFiGAN）

[63] H. Liu et al., "VoiceFixer: A unified framework for high-fidelity speech restoration," Interspeech 2022.
[59] R. Mira et al., "LA-VocE: Low-SNR audio-visual speech enhancement using neural vocoders," ICASSP 2023.

生成式语音增强方法：基于 VAE 的语音增强方法

Method	Year	VAE Var.	Speech Prior	Noise Prior	Infer.
Bando et al.[27]	2018	VAE	VAE	NMF	MCMC
Leglaive et al.[51]	2018	VAE	VAE	NMF	MCEM
RVAE[28]	2020	RVAE	RVAE	NMF	VEM
Guided VAE[64]	2021	CVAE	VAE	NMF	MCEM
NA-VAE[65]	2021	CVAE	VAE	NMF	MCEM
DVAE-VEM[29]	2022	DVAE	DVAE	NMF	VEM
Bayes-PVAE[66]	2022	PVAE	C-VAE	N-VAE	VAE
DCCRN-VAE[67]	2024	Complex VAE	C-VAE	N-VAE	VAE
St-VAE[68]	2024	Weighted VAE	VAE	NMF	VEM
VAE-GAN[69]	2024	β-VAE + GAN	β-PVAE	β-PVAE	VAE

- 基本框架：VAE 建模纯净语音，NMF 建模噪声，贝叶斯推断构成概率 Wiener 滤波
- 改进方法：引入条件 VAE 与语音/噪声独立 VAE，Bayes-PVAE、DCCRN-VAE
- 最新进展：St-VAE 加权方差、VAE-GAN（β-VAE ＋对抗）提升语音质量

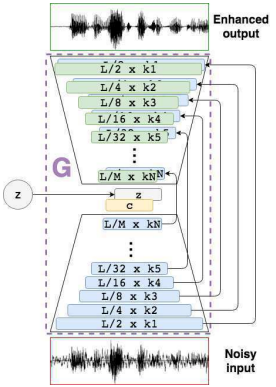
[27] Y. Bando et al., "Statistical speech enhancement based on probabilistic integration of variational autoencoder and non-negative matrix factorization," ICASSP 2018.
[29] X. Bie et al., "Unsupervised speech enhancement using dynamical variational autoencoders," IEEE/ACM TASLP, 2022.

生成式语音增强方法：基于 GAN 的语音增强方法

Method	Year	Generator	Discriminator
SEGAN[24]	2017	U-Net	Waveform T/F
FC-GAN[70]	2018	DNN	Mask / STFT T/F
MMSE-GAN[71]	2018	DNN	Spectral T/F
UNetGAN[72]	2019	U-Net	Waveform T/F
SERGAN[73]	2019	U-Net	Waveform T/F
S-ForkGAN[74]	2019	Forked Gen.	Log-Power Spectral T/F
CP-GAN[75]	2020	Dense Pyramid	Waveform T/F
PAGAN[76]	2020	CNN+LSTM	Waveform T/F
MOCycleGAN[77]	2020	MLP	Mag./Log-Power Spectral T/F
NAT-SE[78]	2021	TCN	Noise-type classifier
GAN-in-GAN[79]	2023	Restormer	Metric + Multi-Res T/F

- **核心机制**：通过对抗学习匹配语音分布，有效缓解单点估计带来的过度平滑问题
- **代表工作**：SEGAN 开创性地引入 U-Net 架构与波形域判别器
- **指标驱动**：MetricGAN 以可微的指标预测网络替代传统的真假二分类判别器，实现对感知评价分数（如 PESQ）的直接优化

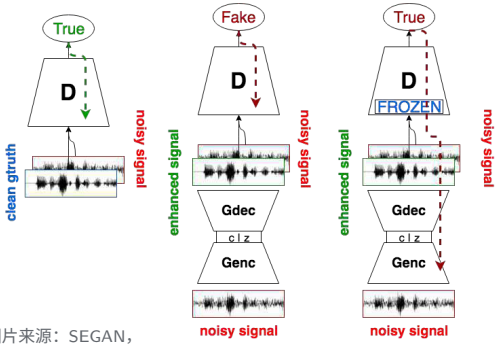
基于 GAN 的语音增强方法：SEGAN



生成器结构：U-Net 结构、中间层加入潜变量 z

$$\min_D V(D) = \frac{1}{2}\mathbb{E}_{\mathbf{x},\mathbf{x}_c}\left[(D(\mathbf{x},\mathbf{x}_c)-1)^2\right] + \frac{1}{2}\mathbb{E}_{\mathbf{z},\mathbf{x}_c}\left[D(G(\mathbf{z},\mathbf{x}_c),\mathbf{x}_c)^2\right]$$

$$\min_G V(G) = \frac{1}{2}\mathbb{E}_{\mathbf{z},\mathbf{x}_c}\left[(D(G(\mathbf{z},\mathbf{x}_c),\mathbf{x}_c)-1)^2\right] + \lambda\|G(\mathbf{z},\mathbf{x}_c)-\mathbf{x}\|_1$$



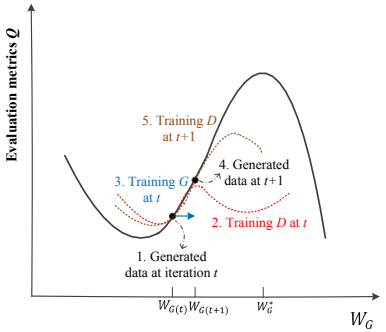
图片来源：SEGAN，

条件 LSGAN：交替训练与重建约束

1. 判别器：真实配对的目标为 1，生成配对为 0
2. 生成器：冻结判别器参数，让生成配对的评分接近 1
3. L_1 重建项：减少增强波形与纯净目标的差异

[24] S. Pascual et al., "SEGAN: Speech enhancement generative adversarial network," Interspeech 2017.

基于 GAN 的语音增强方法：MetricGAN



图片来源：MetricGAN，Fig. 1

判别器 D：拟合归一化指标代理

$$L_D(\text{MetricGAN}) = \mathbb{E}_{x,y}\big[(D(y,y)-1)^2 + \big(D(G(x),y) - Q'(G(x),y)\big)^2\big]$$

$Q' \in [0,1]$ 为归一化指标，训练目标为 1

生成器 G：逼近指定指标分数

$$L_G(\text{MetricGAN}) = \mathbb{E}_x\big[(D(G(x),y)-s)^2\big]$$

s 为目标分数；优化最高指标时取 $s=1$

- 交替更新：D 拟合指标，G 追踪目标分数
- 模型推理时只需 G
- 容易过拟合到 PESQ 分数

[80] S.-W. Fu et al., "MetricGAN: Generative adversarial networks based black-box metric scores optimization for speech enhancement," ICML 2019.
[81] S.-W. Fu et al., "MetricGAN+: An improved version of MetricGAN for speech enhancement," Interspeech 2021.
[82] D. de Oliveira et al., "The PESQetarian: On the relevance of Goodhart's law for speech enhancement," Interspeech 2024.
[52] I. J. Goodfellow et al., "Generative adversarial nets," NIPS 2014.
[24] S. Pascual et al., "SEGAN: Speech enhancement generative adversarial network," Interspeech 2017.

生成式语音增强方法：扩散、得分与流匹配

将语音增强问题转化为“将带噪观测推向纯净分布”的连续概率输运问题：

框架	路径 / 随机量	训练目标（条件 $\mathbf{y}$）	推断方式
去噪扩散	闭式加噪 $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon$	$\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{y})\|_2^2$	反向逐步去噪，T 步
得分 SDE	$d\mathbf{x}_t = f(\mathbf{x}_t, t; \mathbf{y})\,dt + g(t)\,d\boldsymbol{w}_t$	$s_\theta \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t \mid \mathbf{y}) = -\epsilon/\sigma_t$	反向 SDE 或概率流 ODE
流匹配	直线插值 $\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1$	$\|v_\theta(\mathbf{x}_t, t, \mathbf{y}) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2$	沿速度场积分（Euler）

- **采样可确定化**：在精确得分的指导下，反向 SDE 与概率流 ODE 具有相同的边缘分布，意味着扩散生成并非必然伴随随机性
- **直接回归速度场**：流匹配沿着学到的速度场进行积分，训练时无需展开 ODE 求解，推理时随着步数的增大，积分结果收敛到目标分布

核心目标一致（将带噪观测推向纯净分布），差异仅在于监督信号的构建与采样路径的选择

[53] J. Ho et al., “Denoising diffusion probabilistic models,” NeurIPS 2020.
[83] Y. Song et al., “Score-based generative modeling through stochastic differential equations,” ICLR 2021.
[54] Y. Lipman et al., “Flow matching for generative modeling,” ICLR 2023.
[6] J. Richter et al., “Speech enhancement and dereverberation with diffusion-based generative models,” IEEE/ACM TASLP, 2023.

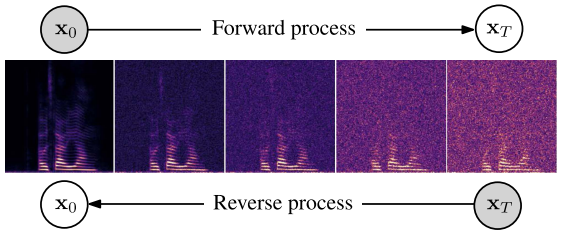
代表性扩散与流匹配语音增强方法

Method	Year	Paradigm	NFEs	Architecture
DiffuSE[84]	2021	Diffusion	50	DiffWave
CDiffuSE[85]	2022	Diffusion	6/50	DiffWave
SGMSE[86]	2022	Score-based	100†	Complex U-Net
SGMSE+[6]	2023	Score-based	60	NCSN++
BBED[87]	2023	Brownian Bridge	60	NCSN++
CRP[88]	2024	Score-based	1–5	NCSN++
LARF[89]	2024	Rectified Flow	1	U-Net
NASE[90]	2024	Score-based	60	NCSN++
UNIVERSE++[91]	2024	Score-based	8	Conditioner + U-Net
SB-SE[92]	2024	Schrödinger Bridge	50	NCSN++
FlowSE-TF[93]	2025	Flow Matching	5	NCSN++
FlowSE-Vocos[25]	2025	Flow Matching	64‡	DiT + Vocos
ROSE-CD[94]	2025	Score-based	1	NCSN++
MeanSE[95]	2026	Mean Flow	1	NCSN++

- 早期扩散借用 DiffWave 声码器，以带噪 Mel 谱为条件生成语音
- SGMSE 系列转向复数 STFT 得分建模
- 更少的推理步数：CRP 预测损失微调、ROSE-CD 一致性蒸馏
- 流匹配以更少推理步数达到相当质量：FlowSE-TF 5 步、MeanSE 均值流单步即可
- FlowSE-Vocos 结合 DiT 预测增强 Mel 谱与 Vocos 波形生成
- 骨干网络：NCSN++、U-Net、DiT 等

[86] S. Welker et al., "Speech enhancement with score-based generative models in the complex STFT domain," Interspeech 2022.
[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.

基于得分的生成式语音增强方法：SGMSE



图片来源：SGMSE+，Fig. 1

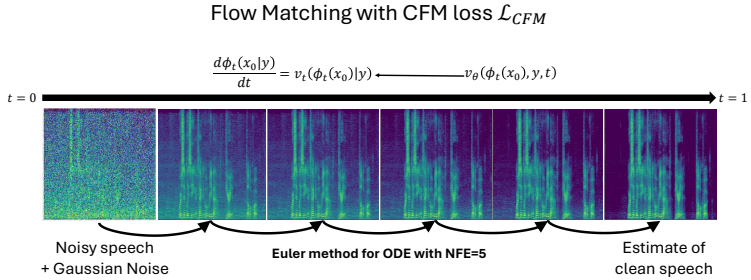
- 复数 STFT 表征，带噪语音作条件
- 前向 SDE 加噪并向观测迁移；得分网络根据 x_t, y, t 估计得分，反向 SDE 迭代恢复
- SGMSE+ 用 NCSN++ 替代 SGMSE 的复数 U-Net 主干

WSJ0-CHiME3 denoising

Method	PESQ↑	SI-SDR↑	SI-SIR↑	DNSMOS↑
Noisy input	2.01	13.5	18.7	3.34
SGMSE[86]	2.28	14.8	25.4	3.70
SGMSE+[6]	2.96	18.3	31.1	3.99
MetricGAN+[81]	3.03	10.5	24.5	3.67
Conv-TasNet[9]	2.99	19.9	29.2	3.79

[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.
[86] S. Welker et al., "Speech enhancement with score-based generative models in the complex STFT domain," Interspeech 2022.

基于流匹配的生成式语音增强方法：FlowSE-TF



图片来源：FlowSE-TF，Fig. 1

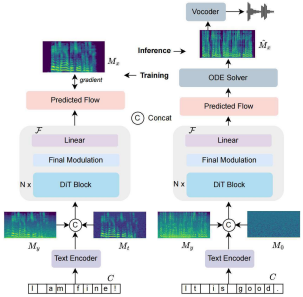
- 带噪语音与高斯扰动构造起点
- 条件流匹配学习时频域速度场
- Euler 沿路径逐步恢复纯净谱

WSJ0–CHiME3，NFE=5（Table II）

训练方法	PESQ↑	WVMOS↑	SI-SDR (dB)↑
Flow matching	3.03	4.00	18.95
Score matching	3.01	3.92	18.57

[93] S. Lee et al., "FlowSE: Flow matching-based speech enhancement," ICASSP 2025.

基于流匹配的生成式语音增强方法：FlowSE-Vocos



图片来源：FlowSE-Vocos，Fig. 1

模型结构：

- 带噪 Mel 谱与可选文本提供条件
- DiT 预测速度，ODE 生成增强 Mel 谱
- Vocos 将增强 Mel 谱合成为波形

DNS Challenge, no reverb

Method	DNSMOS OVRL↑	Spk Sim↑
MaskSR[96]	3.339	0.929
FlowSE, w/o text	3.445	0.934
FlowSE, w/ text	3.451	0.940

[25] Z. Wang et al., "FlowSE: Efficient and high-quality speech enhancement via flow matching," Interspeech 2025.

减少 NFEs：从多步采样到单步生成

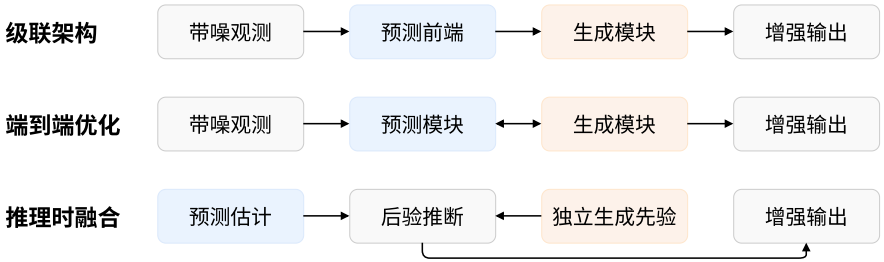
方法	机制	NFEs
SGMSE+[6]	预测–校正采样	60
FlowSE-Vocos[25]	无分类器引导	64‡
FlowSE-TF[93]	少步流匹配 ODE	5
CRP[88]	预测损失微调	1–5
LARF[89]	Rectified Flow	1
ROSE-CD[94]	一致性蒸馏	1
MeanSE[95]	条件均值流	1

- **采样成本**：由 NFE 决定，预测-校正与 CFG 均会使 NFE 翻倍
- **模型蒸馏**：多步教师提供 ODE 轨迹，训练单步学生。如 ROSE-CD 单步 PESQ 3.49，超越 30 步教师模型
- **微调与替代**：CRP 用预测损失微调至 1–5 步；LARF、SFMSE、MeanSE 等均实现了单步生成。
- **实测效果**：FlowSE-TF 以 5 步流匹配 ODE 达到相当的语音质量

最近的研究趋势是：通过蒸馏、微调或替代方法，将多步扩散/流匹配模型减少为 1∼5 步生成模型，从而显著降低推理算法复杂度

[6] J. Richter et al., "Speech enhancement and dereverberation with diffusion-based generative models," IEEE/ACM TASLP, 2023.
[88] B. Lay et al., "Single and few-step diffusion for generative speech enhancement," ICASSP 2024.
[94] L. Xu et al., "Robust one-step speech enhancement via consistency distillation," WASPAA 2025.
[93] S. Lee et al., "FlowSE: Flow matching-based speech enhancement," ICASSP 2025.
[95] J. Wang et al., "MeanSE: Efficient generative speech enhancement with mean flows," ICASSP 2026.

生成式语音增强方法：预测–生成混合式方法



- **级联架构 (Cascaded)**：将系统分解为两个连续阶段，首先由预测式前端产生初始估计或条件信号，随后由生成式后端进行细节优化
- **端到端架构 (End-to-end)**：将预测与生成组件紧密集成在单一的联合优化模型中，从而消除了级联流程中严格的阶段隔离限制
- **推理时融合 / 即插即用**：第三种混合策略不在训练阶段进行联合优化，而是在推理阶段将预测模型与生成先验相结合

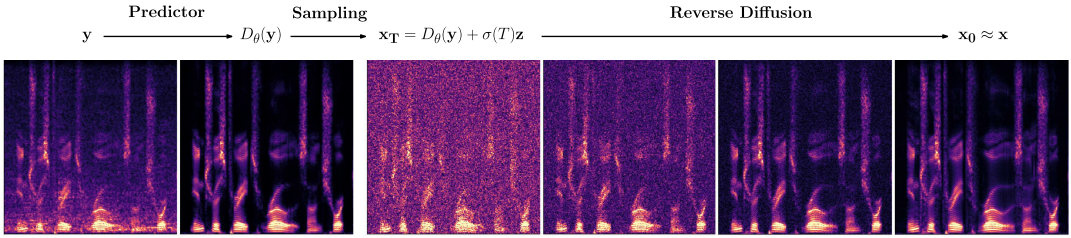
代表性预测–生成混合语音增强方法

Method	Year	Framework	Hybrid Strategy
StoRM[30]	2023	Cascaded	Predictive + Refinement
MOSE[97]	2023		Diffusion + Metric Guidance
CDDSE[98]	2023		Predictive + Diffusion
FUSE[99,100]	2024		Predictive + Diffusion
D2U-SE[101]	2025		Conditional Diffusion
GP-Unified[102]	2024	End-to-End	Fused Predictive + Diffusion
GALD-SE[103]	2024		Mask Guidance + Diffusion
Thunder[104]	2024		Unified Regression–Diffusion
PGUSE[32]	2025		Predictive + Diffusion Branches
HyFlowSE[33]	2026		Flow + Adv. Loss
Diffiner[31]	2023	Plug-and-Play	DDRM Posterior Sampling
SIPS[34]	2026		Predictive Drift + Score Prior

- 预测式高效低延迟，生成式感知更自然；预测-生成混合式方法结合两者优势
- 串联：预测前端抑噪＋生成后端恢复真实感（StoRM、FUSE）
- 端到端：共享表征全程交互、减少误差传播（GP-Unified、Thunder、PGUSE、HyFlowSE）
- 即插即用：独立训练的预测估计＋预训练任务无关先验（Diffiner、SIPS）

[30] J.-M. Lemercier et al., "StoRM: A diffusion-based stochastic regeneration model for speech enhancement and dereverberation," IEEE/ACM TASLP, 2023.
[32] J. Zhang et al., "A composite predictive-generative approach to monaural universal speech enhancement," IEEE TASLP, 2025.

预测–生成混合式生成式方法：StoRM



图片来源：StoRM，Fig. 4

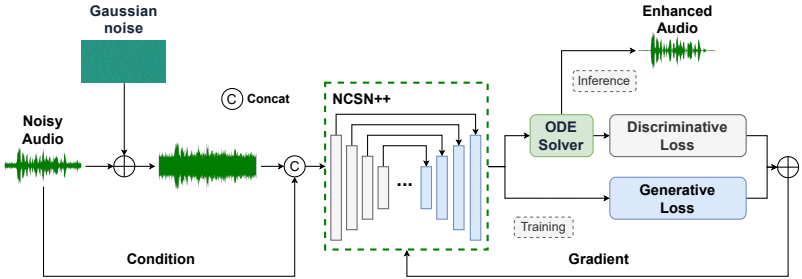
- 预测器先得到纯净语音的初步估计
- 对估计加噪，构造扩散初始状态
- 得分网络结合条件进行逆扩散
- 在语音去噪声和去混响任务上均有提升

WSJ0+CHiME, 50 steps

Condition	PESQ↑	SI-SDR (dB)↑
Noisy	2.30 ± 0.60	11.5 ± 5.2
PostDenoiser	2.50 ± 0.62	14.7 ± 4.3
Both	2.53 ± 0.63	15.1 ± 4.2

[30] J.-M. Lemercier et al., "StoRM: A diffusion-based stochastic regeneration model for speech enhancement and dereverberation," IEEE/ACM TASLP, 2023.

预测–生成混合式生成式方法：HyFlowSE



图片来源：HyFlowSE，Fig. 1

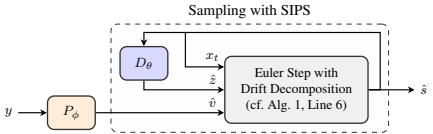
- 流匹配目标学习生成速度场
- 重建损失通过可微 ODE 反传
- 端到端训练，兼顾信号保真与感知质量
- 训练和推理的 ODE 步数绑定

VoiceBank+DEMAND, NFE=5 (Table 1)

Method	Param.(M)	PESQ↑	SI-SDR (dB)↑
FlowSE(M)	27.8	2.98	18.97
CasFlowSE	27.8	3.05	19.13
HyFlowSE(M)	27.8	3.26	19.20
HyFlowSE(T)	5.2	3.21	19.09

[33] Y. Zhang et al., "HyFlowSE: Hybrid end-to-end flow-matching speech enhancement via generative-discriminative learning," ICASSP 2026.

框架：预测漂移与独立语音先验

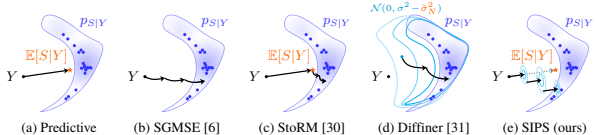


图片来源：SIPS，Fig. 1

$$\widehat{v} = P_\phi(\mathbf{y}) - \mathbf{y}, \qquad \widehat{z} = D_\theta(t, \mathbf{x}_t)$$

- 预测器 P_ϕ 估计纯净语音 $\mathbf{y}$，提供漂移方向
- 去噪器 D_θ 估计当前扰动 $\mathbf{z}_t$，提供采样速度
- 先验仅用纯净语音独立训练，可跨预测器复用

不同方法的条件与耦合位置



图片来源：SIPS，Fig. 2

VoiceBank–DEMAND: matched condition

System	SI-SDR (dB)↑	PESQ↑	NISQA↑	WER (%)↓
SEMamba[49]	19.72	3.56	4.60	8.87
+ StoRM[30]	18.89	3.17	4.54	9.37
+ Diffiner[31]	16.51	2.87	4.81	12.43
+ SIPS	19.63	3.43	4.73	8.81

[34] J. Richter et al., "Predictive-generative drift decomposition for speech enhancement and separation," arXiv:2605.06189, 2026.
[31] R. Sawata et al., "Diffiner: A versatile diffusion-based generative refiner for speech enhancement," Interspeech 2023.

休息 5 分钟

后半程：模型架构、通用语音增强、评测与展望

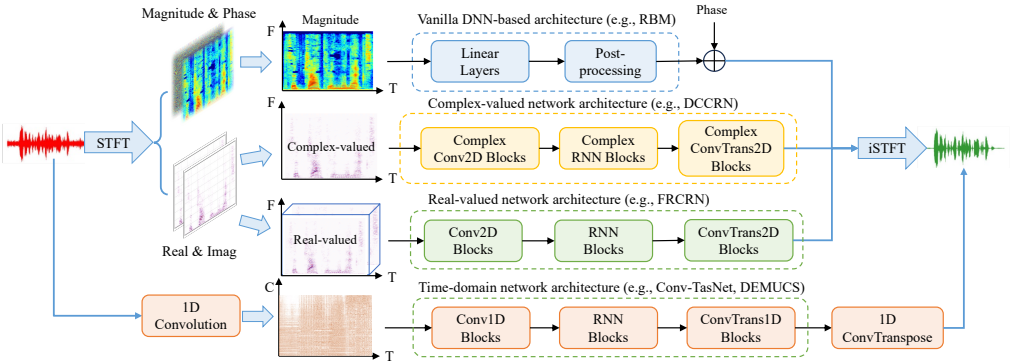
讨论

结合实际应用思考：允许的延迟是多少？哪些信息不可改变？

目录

1. 问题定义与语音表征
2. 语音增强模型的学习策略
3. 建模范式的演进：预测式、生成式与混合框架
4. 语音增强模型的架构演进
5. 迈向通用语音增强
6. 数据集与评测
7. 挑战与未来方向

语音增强模型的架构演进



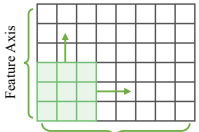
主流语音增强架构大致可分为三大类：

- **时频域实值网络**：例如基于 RBM 的模型
- **时频域复值网络**：例如 DCCRN
- **时域网络**：例如 Conv-TasNet 和 DEMUCS

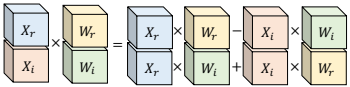
代表性层结构：1D/2D/复数卷积与双路 RNN



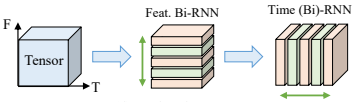
(a) 1D Convolution for raw audio



(b) 2D Convolution for spectrogram



(c) Complex convolution

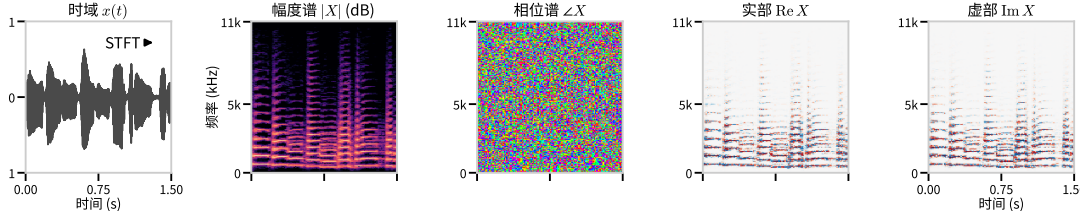


(d) Dual-path RNN

左图展示了各类代表性操作：

- **(a) 1D 卷积**：沿原始音频的时间轴处理波形
- **(b) 2D 卷积**：作用于谱图等时频表示
- **(c) 复数卷积**：联合建模频谱的实部与虚部
- **(d) 双路 RNN**：分块建模帧内与帧间依赖，序列建模更高效
- 卷积、循环与转置卷积模块构成现代语音增强模型的核心组件
- 这些进展推动浅层前馈网络迈向更深的架构，显著提升模型的性能

补充解释：STFT 可以进行幅度、相位或复数谱的处理



$$X = Ae^{j\phi} = X_r + jX_i, \qquad X_r = A\cos\phi, \quad X_i = A\sin\phi$$

处理方式	重建形式	需要解决的问题
幅度增强	$\widehat{A}e^{j\angle X}$	带噪相位误差仍在
幅相分支	$\widehat{A}e^{j\widehat{\phi}}$	相位周期性与一致性
复数估计	$\widehat{X}_r + j\widehat{X}_i$	实虚部耦合与数值稳定性

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.
[43] Z.-Q. Wang et al., "On the compensation between magnitude and phase in speech separation," IEEE Signal Process. Lett., 2021.

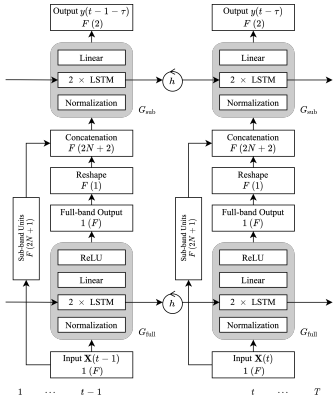
语音增强模型的架构演进

Method	Architecture	Input	Multi-Branch
DNN[36]	FC + Sigmoid	$\log(\mathbf{Y}_m^2)$	✗
CRN[37]	CNN + LSTM	$\mathbf{Y}_m$	✗
Conv-TasNet[9]	1D-CNN	$\mathbf{y}$	✗
DCCRN[5]	Complex NN	$\mathbf{Y}_r + \mathbf{Y}_i$	✗
DEMUCS[44]	1D-CNN + LSTM	$\mathbf{y}$	✗
PHASEN[105]	CNN + BiLSTM	$\mathbf{Y}_r + \mathbf{Y}_i$	✓
FullSubNet[45]	LSTM + Linear	$\mathbf{Y}_m$	✗
DPRNN[106]	1D-CNN + BiRNN	$\mathbf{y}$	✗
CMGAN[48]	CNN + Conformer	$\mathbf{Y}_r+\mathbf{Y}_i+\mathbf{Y}_m$	✓
DeepFilterNet[107]	CNN + Grouped Linear & GRU	$\mathbf{Y}_r+\mathbf{Y}_i+\mathbf{Y}_{\text{ERB}}$	✓
TF-GridNet[16]	CNN + BiLSTM + Self-Atten.	$\mathbf{Y}_r + \mathbf{Y}_i$	✗
MP-SENet[17]	CNN + Conformer	$\mathbf{Y}_m + \mathbf{Y}_p$	✓
NHS-SE+[108]	CNN + LSTM	$\mathbf{Y}_r+\mathbf{Y}_i+\mathbf{Y}_m^e+\mathbf{Y}_m^h$	✓
SEMamba[49]	CNN + Mamba	$\mathbf{Y}_m + \mathbf{Y}_p$	✓
ZipEnhancer[50]	CNN + Zipformer	$\mathbf{Y}_m + \mathbf{Y}_p$	✓

- **多特征融合**：CMGAN 多解码器联合预测幅/实/虚谱；DeepFilterNet 分支处理复频谱与 ERB 频带；NHS-SE+ 联合声道、激励与复数掩码
- **幅相分解**：PHASEN 交互式幅相双分支，MP-SENet 并行解码；该思路延展至 SEMamba、ZipEnhancer 等
- **多分支设计**：从早期的单分支演进为如今普遍采用的显式多分支架构——通过解耦并融合不同语音特征，成为现代语音增强的高效范式

[5] Y. Hu et al., "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," Interspeech 2020.
[17] Y.-X. Lu et al., "Explicit estimation of magnitude and phase spectra in parallel for high-quality speech enhancement," Neural Networks, 2025.

语音增强模型的架构演进：FullSubNet



图片来源：FullSubNet，Fig. 1

用于实时单通道语音增强的全带—子带融合模型：

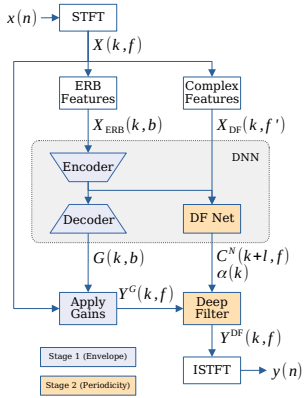
- 全频带模型提取幅度谱的全局特征
- 子带模型融合邻频与全局特征，逐频率估计复数掩码；参数跨频率共享
- 在 DNS Challenge 数据集上的实验表明，FullSubNet 显著优于子带模型、参数量更大的纯全带模型

DNS Challenge test set

	w/ reverb		w/o reverb	
Model	WB-PESQ	SI-SDR	WB-PESQ	SI-SDR
Sub-band	2.650	14.673	2.369	16.153
Full-band	2.681	13.580	2.731	16.190
FullSubNet	2.969	15.750	2.777	17.290

[45] X. Hao et al., "FullSubNet: A full-band and sub-band fusion model for real-time single-channel speech enhancement," ICASSP 2021.

语音增强模型的架构演进：DeepFilterNet



图片来源：DeepFilterNet，Fig. 1

面向全频带（48 kHz）音频的低复杂度语音增强框架，采用两阶段设计：

- 第一阶段：ERB 增益调整感知频带包络
- 第二阶段：复数特征预测深滤波系数，并利用邻近帧信息
- 在 VCTK-DEMAND 测试集上性能与 DCCRN+ 相当但计算复杂度显著更低，适合实时应用

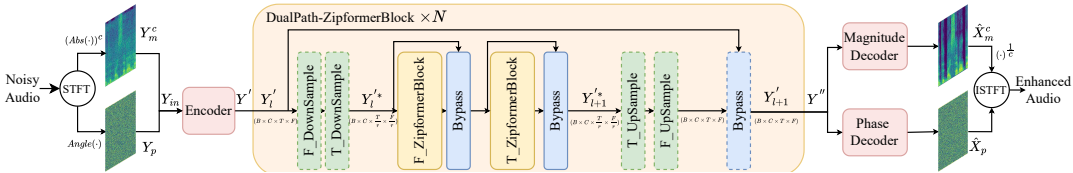
VCTK–DEMAND ablation

Model	Param.(M)	WB-PESQ↑	SI-SDR (dB)↑
DeepFilterNet	1.8	2.81	16.63
w/o stage 2	0.9	2.57	13.81

两阶段输出按帧间周期性加权融合。

[107] H. Schroter et al., "DeepFilterNet: A low complexity speech enhancement framework for full-band audio based on deep filtering," ICASSP 2022.

语音增强模型的架构演进：ZipEnhancer



图片来源：ZipEnhancer，Fig. 1

- **宏观架构**：在时间与频率两个维度上采用对称的下采样与上采样操作，有效降低双路径模型的计算开销
- **核心模块**：模型以 **ZipformerBlock** 作为核心组件，实现高效的局部与全局特征提取
- **训练优化**：使用 **ScaleAdam** 优化器与 **Eden** 学习率调度器，进一步提升模型的收敛速度与整体性能

DNS 2020 无混响测试结果

模型	FLOPs (G)	WB-PESQ↑	SI-SDR↑
MP-SENet Up.*	81.33	3.61	20.55
ZipEnhancer(S)	62.85	3.69	21.15
ZipEnhancer(M)	266.96	3.81	22.22

该模型仅用 2.04M 参数和约 62G FLOPS，在公开数据集上分别取得 SOTA 结果

[50] H. Wang et al., "ZipEnhancer: dual-path down-up sampling-based Zipformer for monaural speech enhancement," ICASSP 2025.

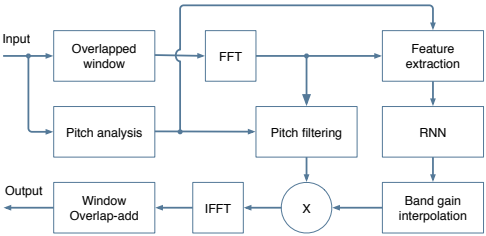
语音增强模型的架构演进：轻量化与实时部署

Method	Architecture	#Params	MACs (M/s)	Latency
RNNoise[109]	GRU	–	–	10 ms
FSPEN[110]	Full-/Sub-band + DPE	79K	89	16.25 ms
SpeechBoost[111]	Wave-U-Net + LSTM	1M	210	3 ms
GTCRN[112]	GT-Conv + G-DPRNN	48.2K	33.9	16 ms
LiSenNet[113]	Conv + Dual-Path Recur.	37K	56	16 ms
Sub-MS[114]	LSTM	626K	–	0.32 ms
SlowFast[115]	Dual-branch + SSM Mod.	–	100	2 ms
UL-UNAS[116]	NAS-based U-Net	170K	35	16 ms
Abs-Hunet[117]	Band-Split + Tri-Fusion HUNet	43.2K	43.2	16 ms

- RNNoise：DSP + GRU 混合框架，88K 参数、10 ms 实现全带实时
- 近期趋势：分组卷积与子带处理——GTCRN 全流程分组，FSPEN 并行子带，LiSenNet 双路径递归，Abs-Hunet 自适应带分裂
- 极低时延：SpeechBoost 面向 TWS 耳机实现 3 ms 时延；Sub-MS 时延进入亚毫秒；SlowFast 以 SSM 节省 70% 计算（时延 62.5 μs）

[112] X. Rong et al., "GTCRN: A speech enhancement model requiring ultralow computational resources," ICASSP 2024.

轻量化与实时部署：RNNoise



图片来源：RNNoise，Fig. 2

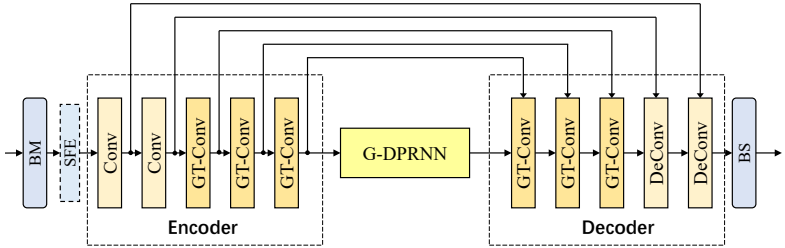
- DSP 端按 Bark 刻度划分 22 个三角频带，全带 48 kHz 实时
- 四层隐藏层（215 单元）的 GRU 网络估计理想临界带增益
- 基音滤波抑制谐波间噪声

Complexity analysis

Item	Value
Network weights	87,503 (8-bit quantizable)
NN compute	17.5 MFlops (real-time)
Total complexity	≈40 MFlops

[109] J.-M. Valin, "A hybrid DSP/deep learning approach to real-time full-band speech enhancement," IEEE MMSP 2018.

轻量化与实时部署：GTCRN



图片来源：GTCRN，Fig. 1

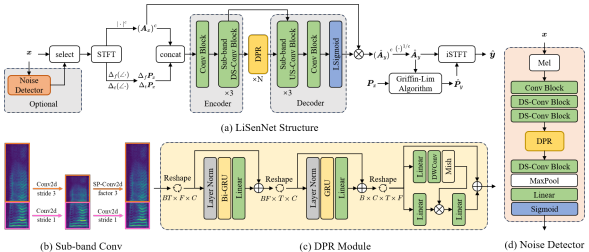
- 编码器通过分组时序卷积提取特征
- 分组双路径循环网络建模时频依赖
- SFE 提取子带特征，TA / TRA 建模时序注意力

DNS3 ablation

Config.	Param.(K)	MACs (M/s)	PESQ↑	SI-SNR↑
Baseline	13.35	33.91	1.87	9.87
+ SFE	15.37	39.07	1.90	10.10
+ SFE + TA	16.86	39.16	1.92	10.29
+ SFE + TRA	23.67	39.63	1.94	10.39

[112] X. Rong et al., "GTCRN: A speech enhancement model requiring ultralow computational resources," ICASSP 2024.

轻量化与实时部署：LiSenNet



图片来源：LiSenNet，Fig. 1

- 通过子带下采样/上采样卷积保留对听觉更重要的低频分辨率
- 用双路径循环模块 (DPR) 配合 ConvGLU 高效建模时频依赖
- 可选噪声检测器按噪声占比自适应增强

VoiceBank–DEMAND testset

Method	Param.	MACs	PESQ↑	STOI↑
Noisy	–	–	1.97	0.921
RNNoise[109]	0.06M	0.04G	2.33	0.922
DCCRN*[5]	3.67M	14.38G	2.78	0.940
FSPEN[110]	79k	89M	2.97	0.942
LiSenNet	37k	56M	3.07	0.939

[113] H. Yan et al., "LiSenNet: Lightweight sub-band and dual-path modeling for real-time speech enhancement," ICASSP 2025.

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迈向通用语音增强

- **现实挑战**：真实场景下的语音常受多种相互交织的退化影响，如混响、削波、带宽限制、编解码伪影和丢包
- **核心目标**：与传统语音增强（聚焦单一退化）不同，通用语音增强旨在从多样且可能复合的失真中恢复高质量语音，且不依赖对失真类型、数量或严重程度的显式假设
- **覆盖范围**：涵盖广泛的损坏类型及其组合包括但不限于：1) 加性噪声；2) 卷积噪声（如混响和回声）；3) 伦巴德效应（Lombard effect）；4) 频谱滤波（如频带限制和均衡）；5) 削波；6) 编解码失真（如 MP3 有损压缩）；7) 不连续传输（如丢包）；8) 处理伪影（如先前算法产生的输出）

[118] J. Serrà et al., "Universal speech enhancement with score-based diffusion," arXiv:2206.03065, 2022.
[119] C. Li et al., "ICASSP 2026 URGENT speech enhancement challenge," ICASSP 2026.
[11] W.-N. Hsu et al., "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," IEEE/ACM TASLP, 2021.
[13] A. Radford et al., "Robust speech recognition via large-scale weak supervision," ICML 2023.
[120] N. Babaev et al., "FINALLY: fast and universal speech enhancement with studio-like quality," NeurIPS 2024.

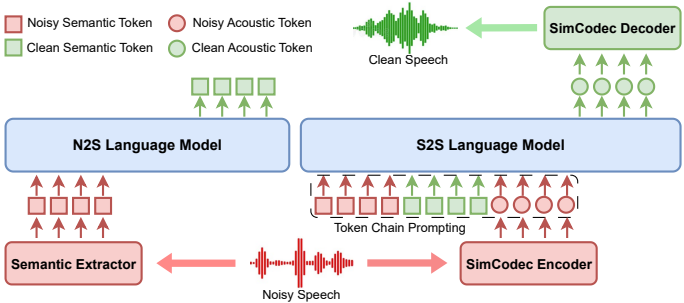
代表性通用语音增强方法

Method	Type	Framework	LLM
VoiceFixer[63]	P	ResUNet + TFGAN	✗
USES[121]	P	TFPSNet + TAC	✗
UNIVERSE++[91]	G	Cond. + Diffusion U-Net	✗
uSee[122]	G	HuBERT + T5 + Diff U-Net	✗
SpeechX[123]	G	EnCodec + VALL-E	✗
FINALLY[120]	G	WavLM + HiFi++	✗
MaskSR[96]	G	DAC + T-F Enc. + MaskGIT	✓
AnyEnhance[124]	G	DAC + w2v-BERT + MaskGIT	✗
PGUSE[32]	P+G	Dual-Branch Pred. + Diff.	✗
GenSE[125]	G	SimCodec + N2S/S2S LMs	✓
LLaSE-G1[126]	G	WavLM + LLaMA + Xcodec2	✓
USEMamba[127]	P+G	TF-Mamba + Flow Matching	✗
SenSE[26]	G	Whisper + LLaMA + DiT + Vocos	✓
Hybrid DGUSE[128]	P+G	TF-GridNet + WavLM + XCodec + LM	✓

- **预测式（早期）**：多条件训练 + 多级处理；严重失真下易过平滑/过抑制
- **生成式方法**：如 VoiceFixer（预测 Mel 谱 + GAN 声码器）、UNIVERSE++（条件扩散），以及引入 SSL 特征的 uSee
- **离散 Token 路线**：以 SpeechX、MaskSR、GenSE、LLaSE-G1 为代表，结合 LLM 进行自回归或掩码建模
- **近期融合趋势**：如 SenSE、Geneses，将 Whisper 编码器、扩散 Transformer 与 LLM 相融合，联合利用声学与语义先验
- **规模法则（Scaling Laws）**：数据多样性与规模的提升，能显著增强对未见声学条件的泛化能力

[91] R. Scheibler et al., "Universal score-based speech enhancement with high content preservation," Interspeech 2024.
[26] X. Li et al., "SenSE: Semantic-aware high-fidelity universal speech enhancement," arXiv:2509.24708, 2025.

通用语音增强方法：GenSE



图片来源：GenSE，Fig. 1

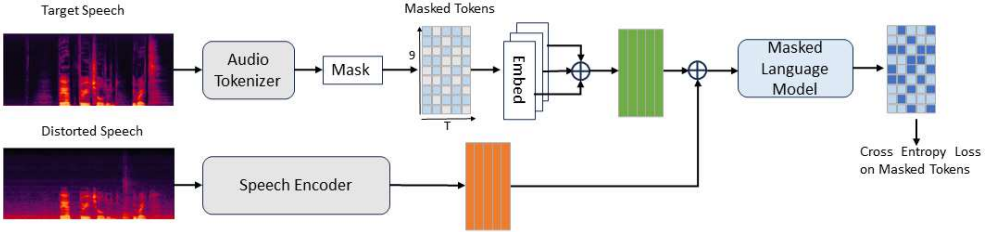
- 基于语言模型的两阶段语音增强框架
- 先用 LM 把带噪语音转成干净语义 token
- 在此基础上生成干净声学 token，并用 SimCodec 重建时域语音

层次框架消融实验

模型/消融	DNSMOS↑	SECS↑	VQ↑
GenSE	3.31	0.66	0.649
w/o token chain	3.28	0.43	0.636
w/o 层次及链提示	3.17	0.41	0.621

[125] J. Yao et al., "GenSE: Generative speech enhancement via language models using hierarchical modeling," ICLR 2025.

通用语音增强方法：MaskSR



图片来源：MaskSR，Fig. 1

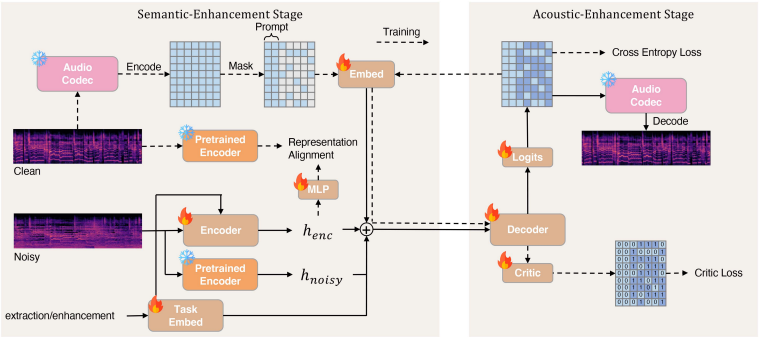
- 基于掩码语言模型的全带宽 44.1 kHz 语音修复系统（噪声、混响、削波和频带限制）
- 用预训练神经编解码器将语音离散化为 token
- 训练时以受损语音为条件预测被掩码的目标 token，推理时通过高效迭代采样重建语音

44.1 kHz SR 带宽扩展

模型	DNSMOS↑	LSD↓	Spk Sim↑
SoundStorm	3.117	1.062	0.812
MaskSR-S	3.135	0.978	0.822
MaskSR-M	3.136	0.959	0.832

[96] X. Li et al., "MaskSR: Masked language model for full-band speech restoration," Interspeech 2024.

通用语音增强方法：AnyEnhance



图片来源：AnyEnhance，Fig. 1

- 基于掩码生成模型的语音增强框架
- 引入提示引导机制和自评机制
- 多个语音与歌声增强任务上均优于单任务模型

Librivox GSR comparison

Spk. prompt	DNSMOS↑	NISQA↑	Similarity↑
w/o	3.308	4.346	0.955
w/	3.401	4.472	0.963

[124] J. Zhang et al., "AnyEnhance: A unified generative model with prompt-guidance and self-critic for voice enhancement," IEEE TASLP, 2025.

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常用语音增强基准数据集概览

杭州电子科技大学 HANGZHOU DIANZI UNIVERSITY

Dataset	Year	Type	Distortions	Lang.	Ch.	Rate
VoiceBank + DEMAND[129]	2016	Syn.	Noise	En.	1	48k
DNS Challenge[130,131]	2020–23	Syn.+Real	Noise, reverb	Multi	1	16–48k
CHiME-3/4[132,133]	2015/16	Syn.+Real	Noise, reverb	En.	1,2,6	16k
CHiME-8 Task 2[134]	2024	Syn.+Real	Noise, reverb, overlap	En.	1,7	16k
WSJ0-2/3mix[135]	2016	Syn.	Speech overlap	En.	1	8/16k
WHAM!/WHAMR![136,137]	2019/20	Syn.	Overlap, noise, reverb	En.	1	8/16k
REVERB Challenge[138]	2014	Syn.+Real	Noise, reverb	En.	1,2,8	16k
AEC Challenge[139,140]	2021–23	Syn.+Real	Echo, noise, reverb	Multi	1,2	16–48k
EARS-WHAM[141]	2024	Syn.	Noise	En.	1	48k
AISHELL-5[142]	2025	Real	Noise, reverb, overlap	Zh.	1,2,4	16/48k
URGENT Challenge[143]	2024–26	Syn.+Real	Noise, reverb, clip, BW, codec/packet loss, wind	Multi	1	8–48k

- VoiceBank + DEMAND：广泛使用的基线数据集；DNS 系列：大规模合成＋真实的噪声抑制基准
- CHiME 系列：从模拟噪声环境到复杂真实场景
- 专用基准：REVERB（去混响）、AEC（回声消除）、AISHELL-5（车内中文）；URGENT 覆盖削波、带宽、编解码/丢包等多样失真

[129] C. Valentini-Botinhao et al., "Investigating RNN-based speech enhancement methods for noise-robust text-to-speech," SSW 2016.
[119] C. Li et al., "ICASSP 2026 URGENT speech enhancement challenge," ICASSP 2026.

常用纯净语音、噪声与 RIR 数据集概览

Dataset	Year	语言/通道	采样率	时长/数量
Clean speech				
WSJ0[144]	1993	English	16 kHz	141 h
VCTK[145]	2016	English	48 kHz	44 h
LibriSpeech[146]	2015	English	16 kHz	1,000 h
GigaSpeech[147]	2021	English	16 kHz	10,000 h
EARS[141]	2024	English	48 kHz	100 h
AISHELL-3[148]	2020	Chinese	44.1 kHz	85 h
Noise				
NOISEX-92[149]	1992	1	16 kHz	1 h
DEMAND[150]	2013	16	48 kHz	1.5 h
AudioSet[151]	2017	1	44.1∼48 kHz	5,800 h
FSD50K[152]	2021	1	44.1 kHz	108.3 h
RIR				
OpenSLR26/28[153]	2017	Synthetic + Real; 1, 2	{8, 16} kHz	1K+
TAU-SRIR[154,155]	2022	Real; 4	24 kHz	38,530
GWA[156]	2022	Synthetic; 1	48 kHz	2M
HARP[157]	2025	Synthetic; 8, 64	48 kHz	100K

- **海量原始数据**：可以由纯净语音、噪声与 RIR 三类源数据混合构建海量训练与评测集；其中，多通道数据可抽取单通道，用于单通道增强任务
- **纯净语音（目标信号）**：提供训练与评测的参考目标，早早期为 WSJ0 小规模数据集，现在为数万小时的大规模语音数据集，中文数据集以 AISHELL 系列为典型代表
- **噪声**：噪声提供背景干扰（如 NOISEX-92/DEMAND 真实噪声，及 AudioSet 5,800 小时弱标注数据）
- **RIR 混响**：RIR 用于编码远场混响特性（如 OpenSLR26/28 真实与合成数据，及 GWA/HARP 大规模合成数据）

语音增强方法评测指标

- **双维度分类**：语音增强的评估指标可沿两个正交维度进行划分——客观/主观，以及侵入式/非侵入式。
- **功能性分类**：按指标的功能角色，还可将其分为三类：
  - **信号级失真**：SNR、Segmental SNR、SI-SNR
  - **感知质量与可懂度**：PESQ、eSTOI、MOS、DNSMOS、UTMOS
  - **下游任务性能**：WER、CER、SpkSim、SpeechBERTScore
- 每种指标各有侧重，评测时应按应用需求选择合适的指标组合

`https://github.com/jiang-wenbin/Eval_SE`

语音增强评测指标汇总

Metric	Obj.	Intr.	Range	Rate
Signal-level distortion				
SNR[1] ↑	✓	✓	$-\infty \sim +\infty$	Any
Segmental SNR[158] ↑	✓	✓	$-\infty \sim +\infty$	Any
SDR / SIR / SAR[159] ↑	✓	✓	$-\infty \sim +\infty$	Any
SI-SDR / SI-SNR[41] ↑	✓	✓	$-\infty \sim +\infty$	Any
LLR[160] ↓	✓	✓	$0 \sim +\infty$	Any
LLSD[161] ↓	✓	✓	$0 \sim +\infty$	Any
MCD[162] ↓	✓	✓	$0 \sim +\infty$	Any
Perceptual quality & intelligibility				
PESQ[163] ↑	✓	✓	$-0.5 \sim 4.5$	8/16k
CSIG / CBAK / COVL[164] ↑	✓	✓	$1 \sim 5$	8/16k
POLQA[165] ↑	✓	✓	$1 \sim 5$	8–48k
MOS[166] ↑	✗	✗	$1 \sim 5$	Any
CMOS[166] ↑	✗	✗	$-3 \sim 3$	Any
DNSMOS[167] P.835 ↑	✓	✗	$1 \sim 5$	16k

Metric	Obj.	Intr.	Range	Rate
Perceptual quality & intelligibility (续)				
UTMOS[168] ↑	✓	✗	$1 \sim 5$	16k
NISQA[169] ↑	✓	✗	$1 \sim 5$	Any
SQUIM (PESQ, STOI, SI-SDR) [170] ↑	✓	✗	–	16k
VQScore[171] ↑	✓	✗	$-1 \sim 1$	16k
SCOREQ[172] ↑	✓	✗	$1 \sim 5$	16 kHz
HASPI[173] ↑	✓	✓	$0 \sim 1$	24 kHz
STOI[174] / eSTOI[175] ↑	✓	✓	$0 \sim 1$	10k
Downstream tasks				
WER[176] / CER[177] ↓	✓	✗	$0 \sim +\infty$	–
SpkSim[143] ↑	✓	✓	$-1 \sim 1$	16k
SpeechBERTScore[178] ↑	✓	✓	$-1 \sim 1$	16k
PhnSim / LPS[177] ↑	✓	✓	$0 \sim 1$	16k

- **多维评估**：单一指标无法全面刻画 SE 性能，需综合信号失真、感知质量与下游任务三类互补指标
- **感知质量**：客观指标分为侵入式（PESQ、CSIG、STOI）与非侵入式（DNSMOS、UTMOS）；主观评估则以人工听音 MOS 为主
- **下游任务**：WER/CER 衡量内容可懂度；SpkSim 与 SpeechBERTScore 评估声纹与语义信息的保留程度
- **评测工具**：可借鉴 VERSA[179] 等工具以及本综述论文提供的脚本，实现全面、可复现的评测

[179] J. Shi et al., "VERSA: A versatile evaluation toolkit for speech, audio, and music," NAACL-HLT (System Demonstrations) 2025.

VoiceBank + DEMAND 数据集代表性方法评测结果

Paradigm	Method	Param. (M)	GMACs/s	PESQ↑	SI-SNR↑ (dB)	UTMOS↑	P.808↑	SpkSim↑	Speech BERTScore↑
Predictive	CMGAN	1.83	31.67	3.41	20.22	4.04	3.54	0.83	0.94
	GTCRN[112]	0.05	0.03	2.87	18.83	3.83	3.45	0.89	0.91
	SEMamba[49]	2.25	32.75	3.55	20.20	4.14	3.61	0.92	0.94
	ZipEnhancer[50]	2.04	41.24	3.63	15.57	4.06	3.54	0.93	0.93
Generative	SGMSE+	65.60	8019.22	2.93	17.50	3.97	3.52	0.86	0.89
	CRP[88]	65.60	668.27	3.08	19.68	4.01	3.57	0.80	0.89
	FlowSE-TF[93]	65.60	668.27	3.12	18.95	4.00	3.56	0.81	0.92
	SenSE[26]	1575.74	431.61	2.75	-8.62	3.97	3.68	0.72	0.88
Hybrid	StoRM	55.12	15905.02	2.93	18.52	3.92	3.54	0.79	0.90
	PGUSE[32]	5.14	25.70	3.24	18.00	3.92	3.48	0.91	0.93
	HyFlowSE[33]	5.20	49.06	3.21	19.12	4.01	3.52	0.92	0.91
	SIPS[34]	64.72	2033.29	3.41	19.63	4.09	3.56	0.91	0.93

- 部分采用基于声码器合成的方法，其评估得到的 SI-SNR 值为负，但感知质量指标仍然较高
- 可能源于声码器的相位或局部时序改变，导致输出波形与参考波形差异较大，但听感仍然自然

测评结果揭示了在重建保真度、感知质量、下游信息保留与计算效率之间的权衡：

- **预测式方法**：在保真度与效率之间取得了较佳平衡，通常能可靠地保留说话人与语言信息
- **生成式方法**：感知质量出色，但信号层面与下游性能在系统间差异较大；此外，迭代采样会显著增加计算成本
- **混合式方法**：通过结合预测估计与生成建模提供灵活折中，但其实际效率高度依赖于具体的生成式机制
- **评价维度的局限性**：单一指标无法全面刻画语音增强算法的性能，需要多指标综合测评

实际应用场景，需要综合考量信号级保真度、感知质量、下游任务性能与计算复杂度

目录

1. 问题定义与语音表征
2. 语音增强模型的学习策略
3. 建模范式的演进：预测式、生成式与混合框架
4. 语音增强模型的架构演进
5. 迈向通用语音增强
6. 数据集与评测
7. 挑战与未来方向

- **低延迟与高效推理**
  - **挑战**：生成式模型迭代采样算法复杂度高；少步/单步生成的质量与效率很难兼顾
  - **方向**：协同设计高效网络架构与紧凑的生成范式，从而降低模型大小和计算复杂度
- **感知质量与可靠评估**
  - **挑战**：客观指标与主观听感存在脱节；现有测试方法难以有效评估生成式方法的效果；生成模型在内容保真与感知质量之间难以平衡
  - **方向**：建立更贴合人类听觉的评估框架，设计能兼顾内容保真与听感自然度的优化目标
- **鲁棒性与通用泛化**
  - **挑战**：未见噪声场景性能下降，引入的信号失真导致下游任务（如 ASR）性能欠佳
  - **方向**：引入课程学习、元学习及自监督学习等多种方法，缩小与全监督系统的差距
- **与语音基础大模型融合**
  - **挑战**：离散 Token 丢失细粒度声学细节，自回归推理成本高，大模型存在“幻觉”风险
  - **方向**：高保真声学编解码器、流式/非自回归架构，实现任务感知的深度融合

总结与展望：从预测试估计迈向生成式建模

- **统一框架与架构演进**：
  - 重点探讨了**预测式与生成式范式**的融合，并回顾了问题建模与语音表征
  - 梳理了多分支架构与面向实时应用的**高效设计**，以及旨在提升复杂声学条件泛化性的**通用语音增强（Universal SE）**
- **数据集与评估协议**：
  - 总结了代表性数据集与测试标准，强调评估需具备**一致性**与**感知意义**
- **核心挑战与未来方向**：
  - **高效推理**与**感知一致性**
  - 跨复杂条件的**鲁棒性**
  - 语音增强与**大规模语音基础模型**的深度融合

谢谢！

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https://github.com/jiang-wenbin/Eval_SE

讨论

你的任务最需要解决哪一个问题：保真、感知、泛化，还是延迟？

参考文献 I

[1] P. C. Loizou, *Speech Enhancement: Theory and Practice*, 2nd ed. CRC Press, 2013.

[2] E. Vincent, T. Virtanen, and S. Gannot, Eds., *Audio Source Separation and Speech Enhancement*. John Wiley & Sons, 2018.

[3] J.-M. Lemercier, J. Richter, S. Welker, and T. Gerkmann, "Analysing diffusion-based generative approaches versus discriminative approaches for speech restoration," in *Proc. ICASSP*, 2023, pp. 1–5.

[4] J.-M. Lemercier, J. Richter, S. Welker, E. Moliner, V. Välimäki, and T. Gerkmann, "Diffusion models for audio restoration: A review," *IEEE Signal Processing Magazine*, vol. 41, no. 6, pp. 72–84, 2024.

[5] Y. Hu, Y. Liu, S. Lv, M. Xing, S. Zhang, Y. Fu, J. Wu, B. Zhang, and L. Xie, "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," in *Proc. Interspeech*, 2020, pp. 2472–2476.

[6] J. Richter, S. Welker, J.-M. Lemercier, B. Lay, and T. Gerkmann, "Speech enhancement and dereverberation with diffusion-based generative models," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 2351–2364, 2023.

[7] E. Tzinis, Y. Adi, V. K. Ithapu, B. Xu, P. Smaragdis, and A. Kumar, "RemixIT: Continual self-training of speech enhancement models via bootstrapped remixing," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 6, pp. 1329–1341, 2022.

[8] S. S. Stevens, J. Volkmann, and E. B. Newman, "A scale for the measurement of the psychological magnitude pitch," *J. Acoust. Soc. Am.*, vol. 8, no. 3, pp. 185–190, 1937.

[9] Y. Luo and N. Mesgarani, "Conv-TasNet: Surpassing ideal time–frequency magnitude masking for speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 27, no. 8, pp. 1256–1266, 2019.

[10] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in *Proc. ICLR*, 2014.

[11] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 3451–3460, 2021.

[12] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, X. Yu, and F. Wei, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE J. Sel. Top. Signal Process.*, vol. 16, no. 6, pp. 1505–1518, 2022.

[13] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023, pp. 28 492–28 518.

[14] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning," in *Proc. NIPS*, vol. 30, 2017, pp. 6306–6315.

参考文献 II

[15] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, "High fidelity neural audio compression," *Trans. Mach. Learn. Res.*, 2023.

[16] Z.-Q. Wang, S. Cornell, S. Choi, Y. Lee, B.-Y. Kim, and S. Watanabe, "TF-GridNet: Integrating full-and sub-band modeling for speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 3221–3236, 2023.

[17] Y.-X. Lu, Y. Ai, and Z.-H. Ling, "Explicit estimation of magnitude and phase spectra in parallel for high-quality speech enhancement," *Neural Networks*, vol. 189, p. 107562, 2025.

[18] M. M. Kashyap, A. Tambwekar, K. Manohara, and S. Natarajan, "Speech denoising without clean training data: A Noise2Noise approach," in *Proc. Interspeech*, 2021, pp. 2716–2720.

[19] Y. Zhang, W. Jiang, Q. Zhuo, and K. Yu, "Iterative noisy-target approach: Speech enhancement without clean speech," in *Proc. NCMMSC*, 2024, pp. 256–264.

[20] W. Jiang, F. Wen, and K. Yu, "SelfSE: Self-supervised speech enhancement via noisy speech refinement," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 34, pp. 2115–2127, 2026.

[21] S.-W. Fu, C. Yu, K.-H. Hung, M. Ravanelli, and Y. Tsao, "MetricGAN-U: Unsupervised speech enhancement/dereverberation based only on noisy/reverberated speech," in *Proc. ICASSP*, 2022, pp. 7412–7416.

[22] H.-Y. Lin, H.-H. Tseng, X. Lu, and Y. Tsao, "Unsupervised noise adaptive speech enhancement by discriminator-constrained optimal transport," in *Proc. NeurIPS*, vol. 34, 2021, pp. 19 935–19 946.

[23] W. Jiang, K. Yu, and F. Wen, "Unsupervised speech enhancement using optimal transport and speech presence probability," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 4445–4455, 2024.

[24] S. Pascual, A. Bonafonte, and J. Serrà, "SEGAN: Speech enhancement generative adversarial network," in *Proc. Interspeech*, 2017, pp. 3642–3646.

[25] Z. Wang, Z. Liu, X. Zhu, Y. Zhu, M. Liu, J. Chen, L. Xiao, C. Weng, and L. Xie, "FlowSE: Efficient and high-quality speech enhancement via flow matching," in *Proc. Interspeech*, 2025, pp. 4858–4862.

[26] X. Li, H. Xie, Z. Wang, Z. Zhang, L. Xiao, S. Wang, and L. Xie, "SenSE: Semantic-aware high-fidelity universal speech enhancement," *arXiv preprint arXiv:2509.24708*, 2025.

[27] Y. Bando, M. Mimura, K. Itoyama, K. Yoshii, and T. Kawahara, "Statistical speech enhancement based on probabilistic integration of variational autoencoder and non-negative matrix factorization," in *Proc. ICASSP*, 2018, pp. 716–720.

参考文献 III

[28] S. Leglaive, X. Alameda-Pineda, L. Girin, and R. Horaud, "A recurrent variational autoencoder for speech enhancement," in *Proc. ICASSP*, 2020, pp. 371–375.

[29] X. Bie, S. Leglaive, X. Alameda-Pineda, and L. Girin, "Unsupervised speech enhancement using dynamical variational autoencoders," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 30, pp. 2993–3007, 2022.

[30] J.-M. Lemercier, J. Richter, S. Welker, and T. Gerkmann, "StoRM: A diffusion-based stochastic regeneration model for speech enhancement and dereverberation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 2724–2737, 2023.

[31] R. Sawata, N. Murata, Y. Takida, T. Uesaka, T. Shibuya, S. Takahashi, and Y. Mitsufuji, "Diffiner: A versatile diffusion-based generative refiner for speech enhancement," in *Proc. Interspeech*, 2023, pp. 3824–3828.

[32] J. Zhang, H. Yan, and X. Li, "A composite predictive-generative approach to monaural universal speech enhancement," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 33, pp. 2312–2325, 2025.

[33] Y. Zhang, W. Jiang, Z. Wang, K. Wu, W. Zhang, and F. Wen, "HyFlowSE: Hybrid end-to-end flow-matching speech enhancement via generative-discriminative learning," in *Proc. ICASSP*, 2026, pp. 16 177–16 181.

[34] J. Richter, Y. Masuyama, C. Boeddeker, T. Edo, G. Wichern, and J. Le Roux, "Predictive-generative drift decomposition for speech enhancement and separation," *arXiv preprint arXiv:2605.06189*, 2026.

[35] I. Cohen and B. Berdugo, "Speech enhancement for non-stationary noise environments," *Signal Process.*, vol. 81, no. 11, pp. 2403–2418, 2001.

[36] Y. Xu, J. Du, L.-R. Dai, and C.-H. Lee, "A regression approach to speech enhancement based on deep neural networks," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 23, no. 1, pp. 7–19, 2015.

[37] K. Tan and D. Wang, "A convolutional recurrent neural network for real-time speech enhancement," in *Proc. Interspeech*, 2018, pp. 3229–3233.

[38] T. Fujimura, Y. Koizumi, K. Yatabe, and R. Miyazaki, "Noisy-target training: A training strategy for DNN-based speech enhancement without clean speech," in *Proc. EUSIPCO*, Aug. 2021, pp. 436–440.

[39] Y. Wang, A. Narayanan, and D. Wang, "On training targets for supervised speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 22, no. 12, pp. 1849–1858, 2014.

参考文献 IV

[40] D. S. Williamson, Y. Wang, and D. Wang, "Complex ratio masking for monaural speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 24, no. 3, pp. 483–492, 2016.

[41] J. Le Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, "SDR–half-baked or well done?" in *Proc. ICASSP*, 2019, pp. 626–630.

[42] J. Engel, L. Hantrakul, C. Gu, and A. Roberts, "DDSP: Differentiable digital signal processing," in *Proc. ICLR*, 2020.

[43] Z.-Q. Wang, G. Wichern, and J. Le Roux, "On the compensation between magnitude and phase in speech separation," *IEEE Signal Process. Lett.*, vol. 28, pp. 2018–2022, 2021.

[44] A. Défossez, G. Synnaeve, and Y. Adi, "Real time speech enhancement in the waveform domain," in *Proc. Interspeech*, 2020, pp. 3291–3295.

[45] X. Hao, X. Su, R. Horaud, and X. Li, "FullSubNet: A full-band and sub-band fusion model for real-time single-channel speech enhancement," in *Proc. ICASSP*, 2021, pp. 6633–6637.

[46] F. Dang, H. Chen, and P. Zhang, "DPT-FSNet: Dual-path transformer based full-band and sub-band fusion network for speech enhancement," in *Proc. ICASSP*, 2022, pp. 6857–6861.

[47] S. Zhao, B. Ma, K. N. Watcharasupat, and W.-S. Gan, "FRCRN: Boosting feature representation using frequency recurrence for monaural speech enhancement," in *Proc. ICASSP*, 2022, pp. 9281–9285.

[48] S. Abdulatif, R. Cao, and B. Yang, "CMGAN: Conformer-based metric GAN for monaural speech enhancement," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 2477–2493, 2024.

[49] R. Chao, W.-H. Cheng, M. La Quatra, S. M. Siniscalchi, C.-H. H. Yang, S.-W. Fu, and Y. Tsao, "An investigation of incorporating Mamba for speech enhancement," in *Proc. SLT*, 2024, pp. 302–308.

[50] H. Wang and B. Tian, "ZipEnhancer: dual-path down-up sampling-based Zipformer for monaural speech enhancement," in *Proc. ICASSP*, 2025, pp. 1–5.

[51] S. Leglaive, L. Girin, and R. Horaud, "A variance modeling framework based on variational autoencoders for speech enhancement," in *Proc. IEEE Int. Workshop Mach. Learn. Signal Process. (MLSP)*, 2018, pp. 1–6.

[52] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. NIPS*, vol. 27, 2014, pp. 2672–2680.

[53] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NeurIPS*, vol. 33, 2020, pp. 6840–6851.

参考文献 V

[54] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *Proc. ICLR*, 2023.

[55] S. Maiti and M. I. Mandel, "Speech denoising by parametric resynthesis," in *Proc. ICASSP*, 2019, pp. 6995–6999.

[56] S. Maiti and M. I. Mandel, "Parametric resynthesis with neural vocoders," in *Proc. WASPAA*, 2019, pp. 303–307.

[57] Z. Du, X. Zhang, and J. Han, "A joint framework of denoising autoencoder and generative vocoder for monaural speech enhancement," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 1493–1505, 2020.

[58] M. Strauss and B. Edler, "A flow-based neural network for time domain speech enhancement," in *Proc. ICASSP*, 2021, pp. 5754–5758.

[59] R. Mira, B. Xu, J. Donley, A. Kumar, S. Petridis, V. K. Ithapu, and M. Pantic, "LA-VocE: Low-SNR audio-visual speech enhancement using neural vocoders," in *Proc. ICASSP*, 2023, pp. 1–5.

[60] P. Andreev, A. Alanov, O. Ivanov, and D. Vetrov, "HiFi++: a unified framework for bandwidth extension and speech enhancement," in *Proc. ICASSP*, 2023, pp. 1–5.

[61] X. Sun, H. Dinkel, Y. Niu, L. Wang, J. Zhang, and J. Luan, "Efficient speech enhancement via embeddings from pre-trained generative audioencoders," in *Proc. Interspeech*, 2025, pp. 4848–4852.

[62] H. R. Guimarães, K. Tan, J. Azcarreta, J. M. Alvarez, P. Agrawal, A. Pandey, and B. Xu, "Improving resource-efficient speech enhancement via neural differentiable DSP vocoder refinement," in *Proc. ASRU*, 2025, pp. 1–7.

[63] H. Liu, X. Liu, Q. Kong, Q. Tian, Y. Zhao, D. Wang, C. Huang, and Y. Wang, "VoiceFixer: A unified framework for high-fidelity speech restoration," in *Proc. Interspeech*, 2022, pp. 4232–4236.

[64] G. Carbajal, J. Richter, and T. Gerkmann, "Guided variational autoencoder for speech enhancement with a supervised classifier," in *Proc. ICASSP*, 2021, pp. 681–685.

[65] H. Fang, G. Carbajal, S. Wermter, and T. Gerkmann, "Variational autoencoder for speech enhancement with a noise-aware encoder," in *Proc. ICASSP*, 2021, pp. 676–680.

[66] Y. Xiang, J. L. Højvang, M. H. Rasmussen, and M. G. Christensen, "A Bayesian permutation training deep representation learning method for speech enhancement with variational autoencoder," in *Proc. ICASSP*, 2022, pp. 381–385.

[67] Y. Xiang, J. Tian, X. Hu, X. Xu, and Z. Yin, "A deep representation learning-based speech enhancement method using complex convolution recurrent variational autoencoder," in *Proc. ICASSP*, 2024, pp. 781–785.

参考文献 VI

[68] A. Golmakani, M. Sadeghi, X. Alameda-Pineda, and R. Serizel, "A weighted-variance variational autoencoder model for speech enhancement," in *Proc. ICASSP*, 2024, pp. 12 451–12 455.

[69] Y. Xiang, J. L. Højvang, M. H. Rasmussen, and M. G. Christensen, "A two-stage deep representation learning-based speech enhancement method using variational autoencoder and adversarial training," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 164–177, 2024.

[70] A. Pandey and D. Wang, "On adversarial training and loss functions for speech enhancement," in *Proc. ICASSP*, 2018, pp. 5414–5418.

[71] M. H. Soni, N. Shah, and H. A. Patil, "Time-frequency masking-based speech enhancement using generative adversarial network," in *Proc. ICASSP*, 2018, pp. 5039–5043.

[72] X. Hao, X. Su, Z. Wang, H. Zhang, and Batushiren, "UNetGAN: A robust speech enhancement approach in time domain for extremely low signal-to-noise ratio condition," in *Proc. Interspeech*, 2019, pp. 1786–1790.

[73] D. Baby and S. Verhulst, "SERGAN: Speech enhancement using relativistic generative adversarial networks with gradient penalty," in *Proc. ICASSP*, 2019, pp. 106–110.

[74] J. Lin, S. Niu, Z. Wei, X. Lan, A. J. van Wijngaarden, M. C. Smith, and K.-C. Wang, "Speech enhancement using forked generative adversarial networks with spectral subtraction," in *Proc. Interspeech*, 2019, pp. 3163–3167.

[75] G. Liu, K. Gong, X. Liang, and Z. Chen, "CP-GAN: Context pyramid generative adversarial network for speech enhancement," in *Proc. ICASSP*, 2020, pp. 6624–6628.

[76] P. Li, Z. Jiang, S. Yin, D. Song, P. Ouyang, L. Liu, and S. Wei, "PAGAN: A phase-adapted generative adversarial networks for speech enhancement," in *Proc. ICASSP*, 2020, pp. 6234–6238.

[77] Y. Xiang and C. Bao, "A parallel-data-free speech enhancement method using multi-objective learning cycle-consistent generative adversarial network," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 1826–1838, 2020.

[78] N. Hou, C. Xu, E. S. Chng, and H. Li, "Learning disentangled feature representations for speech enhancement via adversarial training," in *Proc. ICASSP*, 2021, pp. 666–670.

[79] Y. Duan, J. Ren, H. Yu, and X. Jiang, "GAN-in-GAN for monaural speech enhancement," *IEEE Signal Process. Lett.*, vol. 30, pp. 853–857, 2023.

参考文献 VII

[80] S.-W. Fu, C.-F. Liao, Y. Tsao, and S.-D. Lin, "MetricGAN: Generative adversarial networks based black-box metric scores optimization for speech enhancement," in *Proc. ICML*, vol. 97, Jun. 2019, pp. 2031–2041.

[81] S.-W. Fu, C. Yu, T.-A. Hsieh, P. Plantinga, M. Ravanelli, X. Lu, and Y. Tsao, "MetricGAN+: An improved version of MetricGAN for speech enhancement," in *Proc. Interspeech*, 2021, pp. 201–205.

[82] D. de Oliveira, S. Welker, J. Richter, and T. Gerkmann, "The PESQetarian: On the relevance of Goodhart's law for speech enhancement," in *Proc. Interspeech*, 2024, pp. 3854–3858.

[83] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," in *Proc. ICLR*, 2021.

[84] Y.-J. Lu, Y. Tsao, and S. Watanabe, "A study on speech enhancement based on diffusion probabilistic model," in *Proc. APSIPA ASC*. IEEE, 2021, pp. 659–666.

[85] Y.-J. Lu, Z.-Q. Wang, S. Watanabe, A. Richard, C. Yu, and Y. Tsao, "Conditional diffusion probabilistic model for speech enhancement," in *Proc. ICASSP*, 2022, pp. 7402–7406.

[86] S. Welker, J. Richter, and T. Gerkmann, "Speech enhancement with score-based generative models in the complex STFT domain," in *Proc. Interspeech*, 2022, pp. 2928–2932.

[87] B. Lay, S. Welker, J. Richter, and T. Gerkmann, "Reducing the prior mismatch of stochastic differential equations for diffusion-based speech enhancement," in *Proc. Interspeech*, 2023, pp. 3809–3813.

[88] B. Lay, J.-M. Lemercier, J. Richter, and T. Gerkmann, "Single and few-step diffusion for generative speech enhancement," in *Proc. ICASSP*, 2024, pp. 626–630.

[89] Z. Li and N. Inoue, "Locally aligned rectified flow model for speech enhancement towards single-step diffusion," in *Proc. Interspeech*, 2024, pp. 2195–2199.

[90] Y. Hu, C. Chen, R. Li, Q. Zhu, and E. S. Chng, "Noise-aware speech enhancement using diffusion probabilistic model," in *Proc. Interspeech*, 2024, pp. 2225–2229.

[91] R. Scheibler, Y. Fujita, Y. Shirahata, and T. Komatsu, "Universal score-based speech enhancement with high content preservation," in *Proc. Interspeech*, 2024, pp. 1165–1169.

参考文献 VIII

[92] A. Jukić, R. Korostik, J. Balam, and B. Ginsburg, "Schrödinger bridge for generative speech enhancement," in *Proc. Interspeech*, 2024, pp. 1175–1179.

[93] S. Lee, S. Cheong, S. Han, and J. W. Shin, "FlowSE: Flow matching-based speech enhancement," in *Proc. ICASSP*, 2025, pp. 1–5.

[94] L. Xu, L. F. Yan, and W. B. Kleijn, "Robust one-step speech enhancement via consistency distillation," in *Proc. WASPAA*, 2025, pp. 1–5.

[95] J. Wang, H. Wang, W. Wang, L. Yang, C. Li, W. Zhang, L. Tan, and Y. Qian, "MeanSE: Efficient generative speech enhancement with mean flows," in *Proc. ICASSP*, 2026, pp. 16 827–16 831.

[96] X. Li, Q. Wang, and X. Liu, "MaskSR: Masked language model for full-band speech restoration," in *Proc. Interspeech*, 2024, pp. 2275–2279.

[97] C. Chen, Y. Hu, W. Weng, and E. S. Chng, "Metric-oriented speech enhancement using diffusion probabilistic model," in *Proc. ICASSP*, 2023, pp. 1–5.

[98] H. Wang and D. Wang, "Cross-domain diffusion based speech enhancement for very noisy speech," in *Proc. ICASSP*, 2023, pp. 1–5.

[99] Y. Yang, N. Trigoni, and A. Markham, "Pre-training feature guided diffusion model for speech enhancement," in *Proc. Interspeech*, 2024, pp. 1185–1189.

[100] N. Goswami and T. Harada, "FUSE: Universal speech enhancement using multi-stage fusion of sparse compression and token generation models for the URGENT 2025 challenge," in *Proc. Interspeech*, 2025, pp. 883–887.

[101] H. Shi, X. Lu, K. Shimada, and T. Kawahara, "Combining deterministic enhanced conditions with dual-streaming encoding for diffusion-based speech enhancement," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 33, pp. 4253–4266, 2025.

[102] H. Shi, K. Shimada, M. Hirano, T. Shibuya, Y. Koyama, Z. Zhong, S. Takahashi, T. Kawahara, and Y. Mitsufuji, "Diffusion-based speech enhancement with joint generative and predictive decoders," in *Proc. ICASSP*, 2024, pp. 12 951–12 955.

[103] C. Wang, J. Gu, D. Yao, J. Li, and Y. Yan, "GALD-SE: Guided anisotropic lightweight diffusion for efficient speech enhancement," *IEEE Signal Process. Lett.*, vol. 32, pp. 426–430, 2025.

[104] T. Trachu, C. Piansaddhayanon, and E. Chuangsuwanich, "Thunder: Unified regression-diffusion speech enhancement with a single reverse step using Brownian bridge," in *Proc. Interspeech*, 2024, pp. 1180–1184.

[105] D. Yin, C. Luo, Z. Xiong, and W. Zeng, "PHASEN: A phase-and-harmonics-aware speech enhancement network," in *Proc. AAAI*, vol. 34, no. 05, 2020, pp. 9458–9465.

参考文献 IX

[106] Y. Luo, Z. Chen, and T. Yoshioka, "Dual-path RNN: Efficient long sequence modeling for time-domain single-channel speech separation," in *Proc. ICASSP*, 2020, pp. 46–50.

[107] H. Schroter, A. N. Escalante-B, T. Rosenkranz, and A. Maier, "DeepFilterNet: A low complexity speech enhancement framework for full-band audio based on deep filtering," in *Proc. ICASSP*, 2022, pp. 7407–7411.

[108] W. Jiang and K. Yu, "Speech enhancement with integration of neural homomorphic synthesis and spectral masking," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 1758–1770, 2023.

[109] J.-M. Valin, "A hybrid DSP/deep learning approach to real-time full-band speech enhancement," in *Proc. IEEE MMSP*, 2018, pp. 1–5.

[110] L. Yang, W. Liu, R. Meng, G. Lee, S. Baek, and H.-g. Moon, "FSPEN: An ultra-lightweight network for real time speech enhancement," in *Proc. ICASSP*, 2024, pp. 10 671–10 675.

[111] H. Bae, P. Andreev, A. Saginbaev, N. Babaev, W.-J. Lee, H. Sung, and H.-Y. Cho, "Speech boosting: Low-latency live speech enhancement for TWS earbuds," in *Proc. Interspeech*, 2024, pp. 647–651.

[112] X. Rong, T. Sun, X. Zhang, Y. Hu, C. Zhu, and J. Lu, "GTCRN: A speech enhancement model requiring ultralow computational resources," in *Proc. ICASSP*, 2024, pp. 971–975.

[113] H. Yan, J. Zhang, C. Fan, Y. Zhou, and P. Liu, "LiSenNet: Lightweight sub-band and dual-path modeling for real-time speech enhancement," in *Proc. ICASSP*, 2025, pp. 1–5.

[114] A. Dementyev, C. K. A. Reddy, S. Wisdom, N. Chatlani, J. R. Hershey, and R. F. Lyon, "Towards sub-millisecond latency real-time speech enhancement models on hearables," in *Proc. ICASSP*, 2025, pp. 1–5.

[115] L. Cheng, A. Pandey, B. Xu, T. Delbruck, V. K. Ithapu, and S.-C. Liu, "Modulating state space model with SlowFast framework for compute-efficient ultra low-latency speech enhancement," in *Proc. ICASSP*, 2025, pp. 1–5.

[116] X. Rong, L. Yang, D. Wang, Y. Hu, C. Zhu, K. Chen, and J. Lu, "UL-UNAS: Ultra-lightweight U-Nets for real-time speech enhancement via network architecture search," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 34, pp. 1085–1096, 2026.

[117] X. Wang, T. Jiang, X. Lu, X. Jiang, Y. Ma, and J. Sang, "Abs-Hunet: An ultra-lightweight speech enhancement model with adaptive band-split and half-UNet design," in *Proc. ICASSP*, 2026, pp. 19 927–19 931.

[118] J. Serrà, S. Pascual, J. Pons, R. O. Araz, and D. Scaini, "Universal speech enhancement with score-based diffusion," *arXiv preprint arXiv:2206.03065*, 2022.

参考文献 X

[119] C. Li, W. Wang, M. Sach, W. Zhang, K. Saijo, S. Cornell, Y. Fu, Z. Ni, T. Fingscheidt, S. Watanabe, and Y. Qian, "ICASSP 2026 URGENT speech enhancement challenge," in *Proc. ICASSP*, 2026, pp. 21 919–21 921.

[120] N. Babaev, K. Tamogashev, A. Saginbaev, I. Shchekotov, H. Bae, H. Sung, W. Lee, H.-Y. Cho, and P. Andreev, "FINALLY: fast and universal speech enhancement with studio-like quality," in *Proc. NeurIPS*, vol. 37, 2024, pp. 934–965.

[121] W. Zhang, K. Saijo, Z.-Q. Wang, S. Watanabe, and Y. Qian, "Toward universal speech enhancement for diverse input conditions," in *Proc. ASRU*, 2023, pp. 1–6.

[122] M. Yang, C. Zhang, Y. Xu, Z. Xu, H. Wang, B. Raj, and D. Yu, "uSee: Unified speech enhancement and editing with conditional diffusion models," in *Proc. ICASSP*, 2024, pp. 7125–7129.

[123] X. Wang, M. Thakker, Z. Chen, N. Kanda, S. E. Eskimez, S. Chen, M. Tang, S. Liu, J. Li, and T. Yoshioka, "SpeechX: Neural codec language model as a versatile speech transformer," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 3355–3364, 2024.

[124] J. Zhang, J. Yang, Z. Fang, Y. Wang, Z. Zhang, Z. Wang, F. Fan, and Z. Wu, "AnyEnhance: A unified generative model with prompt-guidance and self-critic for voice enhancement," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 33, pp. 3085–3098, 2025.

[125] J. Yao, H. Liu, C. Chen, Y. Hu, E. Chng, and L. Xie, "GenSE: Generative speech enhancement via language models using hierarchical modeling," in *Proc. ICLR*, 2025, pp. 69 229–69 249.

[126] B. Kang, X. Zhu, Z. Zhang, Z. Ye, M. Liu, Z. Wang, Y. Zhu, G. Ma, J. Chen, L. Xiao, C. Weng, W. Xue, and L. Xie, "LLaSE-G1: Incentivizing generalization capability for LLaMA-based speech enhancement," in *Proc. ACL*, 2025, pp. 13 292–13 305.

[127] R. Chao, R. Nasretdinov, Y.-C. F. Wang, A. Jukic, S.-W. Fu, and Y. Tsao, "Universal speech enhancement with regression and generative Mamba," in *Proc. Interspeech*, 2025, pp. 888–892.

[128] Y. Liu, C. Liu, X. Liang, H. Yan, S. Xue, and Z. Xue, "A hybrid discriminative and generative system for universal speech enhancement," in *Proc. ICASSP*, 2026, pp. 21 907–21 909.

[129] C. Valentini-Botinhao, X. Wang, S. Takaki, and J. Yamagishi, "Investigating RNN-based speech enhancement methods for noise-robust text-to-speech," in *Proc. SSW*, 2016, pp. 146–152.

[130] C. K. Reddy, V. Gopal, R. Cutler, E. Beyrami, R. Cheng, H. Dubey, S. Matusevych, R. Aichner, A. Aazami, S. Braun, P. Rana, S. Srinivasan, and J. Gehrke, "The Interspeech 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results," in *Proc. Interspeech*, 2020, pp. 2492–2496.

参考文献 XI

[131] H. Dubey, A. Aazami, V. Gopal, B. Naderi, S. Braun, R. Cutler, A. Ju, M. Zohourian, M. Tang, M. Golestaneh, and R. Aichner, "ICASSP 2023 deep noise suppression challenge," *IEEE Open J. Signal Process.*, vol. 5, pp. 725–737, 2024.

[132] J. Barker, R. Marxer, E. Vincent, and S. Watanabe, "The third 'CHiME' speech separation and recognition challenge: Dataset, task and baselines," in *Proc. ASRU*, 2015, pp. 504–511.

[133] J. Barker, R. Marxer, E. Vincent, and S. Watanabe, "The third 'CHiME' speech separation and recognition challenge: Analysis and outcomes," *Comput. Speech Lang.*, vol. 46, pp. 605–626, 2017.

[134] A. Vinnikov, A. Ivry, A. Hurvitz, I. Abramovski, S. Koubi, I. Gurvich, S. Peer, X. Xiao, B. M. Elizalde, N. Kanda, X. Wang, S. Shaer, S. Yagev, Y. Asher, S. Sivasankaran, Y. Gong, M. Tang, H. Wang, and E. Krupka, "NOTSOFAR-1 challenge: New datasets, baseline, and tasks for distant meeting transcription," in *Proc. Interspeech*, 2024, pp. 5003–5007.

[135] J. R. Hershey, Z. Chen, J. Le Roux, and S. Watanabe, "Deep clustering: Discriminative embeddings for segmentation and separation," in *Proc. ICASSP*, 2016, pp. 31–35.

[136] G. Wichern, J. Antognini, M. Flynn, L. R. Zhu, E. McQuinn, D. Crow, E. Manilow, and J. L. Roux, "WHAM!: Extending speech separation to noisy environments," in *Proc. Interspeech*, 2019, pp. 1368–1372.

[137] M. Maciejewski, G. Wichern, E. McQuinn, and J. Le Roux, "WHAMR!: Noisy and reverberant single-channel speech separation," in *Proc. ICASSP*, 2020, pp. 696–700.

[138] K. Kinoshita, M. Delcroix, S. Gannot, E. A. P. Habets, R. Haeb-Umbach, W. Kellermann, V. Leutnant, R. Maas, T. Nakatani, B. Raj, A. Sehr, and T. Yoshioka, "The REVERB challenge: A benchmark task for reverberation-robust ASR techniques," in *New Era for Robust Speech Recognition: Exploiting Deep Learning*. Springer, 2017, pp. 345–354.

[139] K. Sridhar, R. Cutler, A. Saabas, T. Parnamaa, M. Loide, H. Gamper, S. Braun, R. Aichner, and S. Srinivasan, "ICASSP 2021 acoustic echo cancellation challenge: Datasets, testing framework, and results," in *Proc. ICASSP*, 2021, pp. 151–155.

[140] R. Cutler, A. Saabas, T. Pärnamaa, M. Purin, E. Indenbom, N.-C. Ristea, J. Gužvin, H. Gamper, S. Braun, and R. Aichner, "ICASSP 2023 acoustic echo cancellation challenge," *IEEE Open J. Signal Process.*, vol. 5, pp. 675–685, 2024.

[141] J. Richter, Y.-C. Wu, S. Krenn, S. Welker, B. Lay, S. Watanabe, A. Richard, and T. Gerkmann, "EARS: An anechoic fullband speech dataset benchmarked for speech enhancement and dereverberation," in *Proc. Interspeech*, 2024, pp. 4873–4877.

[142] Y. Dai, H. Wang, X. Li, Z. Zhang, S. Wang, L. Xie, X. Xu, H. Guo, S. Zhang, H. Bu, and W. Chen, "AISHELL-5: The first open-source in-car multi-channel multi-speaker speech dataset for automatic speech diarization and recognition," in *Proc. Interspeech*, 2025, pp. 5493–5497.

[143] W. Zhang, R. Scheibler, K. Saijo, S. Cornell, C. Li, Z. Ni, J. Pirklbauer, M. Sach, S. Watanabe, T. Fingscheidt, and Y. Qian, "URGENT challenge: Universality, robustness, and generalizability for speech enhancement," in *Proc. Interspeech*, 2024, pp. 4868–4872.

[144] D. B. Paul and J. M. Baker, "The design for the Wall Street Journal-based CSR corpus," in *Proc. Speech Nat. Lang. Workshop*, 1992, pp. 357–362.

[145] C. Veaux, J. Yamagishi, and K. MacDonald, "CSTR VCTK Corpus: English multi-speaker corpus for CSTR voice cloning toolkit," 2016.

[146] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: an ASR corpus based on public domain audio books," in *Proc. ICASSP*, 2015, pp. 5206–5210.

[147] G. Chen, S. Chai, G.-B. Wang, J. Du, W.-Q. Zhang, C. Weng, D. Su, D. Povey, J. Trmal, J. Zhang, M. Jin, S. Khudanpur, S. Watanabe, S. Zhao, W. Zou, X. Li, X. Yao, Y. Wang, Z. You, and Z. Yan, "GigaSpeech: An evolving, multi-domain ASR corpus with 10,000 hours of transcribed audio," in *Proc. Interspeech*, 2021, pp. 3670–3674.

[148] Y. Shi, H. Bu, X. Xu, S. Zhang, and M. Li, "AISHELL-3: A multi-speaker Mandarin TTS corpus," in *Proc. Interspeech*, 2021, pp. 2756–2760.

[149] A. Varga and H. J. M. Steeneken, "Assessment for automatic speech recognition: II. NOISEX-92: A database and an experiment to study the effect of additive noise on speech recognition systems," *Speech Commun.*, vol. 12, no. 3, pp. 247–251, 1993.

[150] J. Thiemann, N. Ito, and E. Vincent, "The diverse environments multi-channel acoustic noise database (DEMAND): A database of multichannel environmental noise recordings," in *Proceedings of Meetings on Acoustics*, vol. 19, 2013, p. 035081.

[151] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio Set: An ontology and human-labeled dataset for audio events," in *Proc. ICASSP*, 2017, pp. 776–780.

[152] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, "FSD50K: an open dataset of human-labeled sound events," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 30, pp. 829–852, 2022.

[153] T. Ko, V. Peddinti, D. Povey, M. L. Seltzer, and S. Khudanpur, "A study on data augmentation of reverberant speech for robust speech recognition," in *Proc. ICASSP*, 2017, pp. 5220–5224.

参考文献 XIII

[154] A. Politis, S. Adavanne, and T. Virtanen, "A dataset of reverberant spatial sound scenes with moving sources for sound event localization and detection," in *Proc. DCASE*, 2020, pp. 165–169.

[155] A. Politis, S. Adavanne, and T. Virtanen, "TAU spatial room impulse response database (TAU-SRIR DB)," 2022.

[156] Z. Tang, R. Aralikatti, A. J. Ratnarajah, and D. Manocha, "GWA: A large high-quality acoustic dataset for audio processing," in *ACM SIGGRAPH 2022 Conference Proceedings*, 2022, pp. 1–9.

[157] S. Saini and J. Peissig, "HARP: A large-scale higher-order ambisonic room impulse response dataset," in *Proc. ICASSPW*, 2025, pp. 1–5.

[158] P. Mermelstein, "Evaluation of a segmental SNR measure as an indicator of the quality of ADPCM coded speech," *J. Acoust. Soc. Am.*, vol. 66, no. 6, pp. 1664–1667, 1979.

[159] E. Vincent, R. Gribonval, and C. Fevotte, "Performance measurement in blind audio source separation," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 14, no. 4, pp. 1462–1469, 2006.

[160] R. Crochiere, J. Tribolet, and L. Rabiner, "An interpretation of the log likelihood ratio as a measure of waveform coder performance," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 28, no. 3, pp. 318–323, 1980.

[161] S. R. Quackenbush, T. P. Barnwell, and M. A. Clements, *Objective measures of speech quality*. Prentice Hall, 1988.

[162] R. Kubichek, "Mel-cepstral distance measure for objective speech quality assessment," in *Proc. PACRIM*, vol. 1, 1993, pp. 125–128.

[163] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, "Perceptual evaluation of speech quality (PESQ)-a new method for speech quality assessment of telephone networks and codecs," in *Proc. ICASSP*, vol. 2, 2001, pp. 749–752.

[164] Y. Hu and P. C. Loizou, "Evaluation of objective quality measures for speech enhancement," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 16, no. 1, pp. 229–238, 2008.

[165] Y. Gaoxiong and Z. Wei, "The perceptual objective listening quality assessment algorithm in telecommunication: Introduction of ITU-T new metrics POLQA," in *Proc. ICCC*, 2012, pp. 351–355.

[166] International Telecommunication Union, *Methods for subjective determination of transmission quality*. International Telecommunication Union, 1996, recommendation ITU-T P.800 (08/1996).

[167] C. K. A. Reddy, V. Gopal, and R. Cutler, "DNSMOS P.835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors," in *Proc. ICASSP*, 2022, pp. 886–890.

参考文献 XIV

[168] T. Saeki, D. Xin, W. Nakata, T. Koriyama, S. Takamichi, and H. Saruwatari, "UTMOS: UTokyo-SaruLab system for VoiceMOS challenge 2022," in *Proc. Interspeech*, 2022, pp. 4521–4525.

[169] G. Mittag, B. Naderi, A. Chehadi, and S. Möller, "NISQA: A deep CNN-self-attention model for multidimensional speech quality prediction with crowdsourced datasets," in *Proc. Interspeech*, 2021, pp. 2127–2131.

[170] A. Kumar, K. Tan, Z. Ni, P. Manocha, X. Zhang, E. Henderson, and B. Xu, "TorchAudio-Squim: Reference-less speech quality and intelligibility measures in torchaudio," in *Proc. ICASSP*, 2023, pp. 1–5.

[171] S.-W. Fu, K.-H. Hung, Y. Tsao, and Y.-C. F. Wang, "Self-supervised speech quality estimation and enhancement using only clean speech," in *Proc. ICLR*, 2024, pp. 8538–8559.

[172] A. Ragano, J. Skoglund, and A. Hines, "SCOREQ: Speech quality assessment with contrastive regression," in *Proc. NeurIPS*, vol. 37, 2024, pp. 105 702–105 729.

[173] J. M. Kates and K. H. Arehart, "The hearing-aid speech perception index (HASPI)," *Speech Commun.*, vol. 65, pp. 75–93, 2014.

[174] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, "An algorithm for intelligibility prediction of time–frequency weighted noisy speech," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 19, no. 7, pp. 2125–2136, 2011.

[175] J. Jensen and C. H. Taal, "An algorithm for predicting the intelligibility of speech masked by modulated noise maskers," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 24, no. 11, pp. 2009–2022, 2016.

[176] A. C. Morris, V. Maier, and P. Green, "From WER and RIL to MER and WIL: improved evaluation measures for connected speech recognition," in *Proc. Interspeech*, 2004, pp. 2765–2768.

[177] J. Pirklbauer, M. Sach, K. Fluyt, W. Tirry, W. Wardah, S. Möller, and T. Fingscheidt, "Evaluation metrics for generative speech enhancement methods: Issues and perspectives," in *Proc. ITG Conference on Speech Communication*, 2023, pp. 265–269.

[178] T. Saeki, S. Maiti, S. Takamichi, S. Watanabe, and H. Saruwatari, "SpeechBERTScore: Reference-aware automatic evaluation of speech generation leveraging NLP evaluation metrics," in *Proc. Interspeech*, 2024, pp. 4943–4947.

[179] J. Shi, H.-j. Shim, J. Tian, S. Arora, H. Wu, D. Petermann, J. Q. Yip, Y. Zhang, Y. Tang, W. Zhang, D. S. Alharthi, Y. Huang, K. Saito, J. Han, Y. Zhao, C. Donahue, and S. Watanabe, "VERSA: A versatile evaluation toolkit for speech, audio, and music," in *Proc. NAACL-HLT (System Demonstrations)*, 2025, pp. 191–209.